ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF DESIGN (RSD)

DEPARTMENT OF INTERIOR DESIGN

COURSE STRUCTURE & SYLLABUS

Bachelor of Interior Design (B.I.D.)

(4-YEARS UNDER GRADUATE PROGRAM)

W.E.F AY: 2026 – 27

COURSE STRUCTURE AND SYLLABUS
DEPARTMENT OF INTERIOR DESIGN

Bachelor of Interior Design
Programme Structure

B.I.D Course Structure (with subject codes)
Course structure and credit summary

Sl. No.	Courses	1st	2nd	3rd	4th	5th	6th	7th	8th	Total Credits
1.	Major	6	6	8	12	12	16	16	4	80
2.	Minor	3	3	4	6	4	4	4	4	32
3.	Interdisciplinary	3	3	3	NA	NA	NA	NA	NA	9
4.	AEC	2	2	2	2	NA	NA	NA	NA	8
5.	SEC	3	3	3	NA	NA	NA	NA	NA	9
6.	VAC (Value added course)	3	3	NA	NA	NA	NA	NA	NA	6
7.	Summer Internship (SI)	NA	NA	NA	NA	NA	NA	4	NA	4
8.	Major Research	NA	NA	NA	NA	NA	NA	NA	12	12
9.	**Total Credits**	**20**	**20**	**20**	**20**	**20**	**20**	**20**	**20**	**160**

Note : There is no multiple entry/exit in the 4 year Bachelor of Interior Design Undergraduate program and students are mandatory to do 8 weeks of Summer/Industry Internship.

1st Semester **Major (Core)**				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M111	Design Thinking	100	3
2	IDS082M112	Basic Design	100	3
Interdisciplinary				
3	IKS992K101	Introduction to Indian Knowledge System – I	100	3
Ability Enhancement Course (AEC)				
4	CEN982A101	Communicative English I	100	2
Skill Enhancement Course (SEC)				
5	IDS082S111	Joinery and Surface Aesthetics	100	3
Value Addition Course (VAC)				
6	VAC992V1413	Indian Architectural Heritage	100	3
SWAYAM COURSE				
7	IDS082Z191	Sustainable Architecture	100	¾
Minor				

8	COD082N111	Illustration Technique	100	3
9	FAS082N114	Fashion Design and Concept	100	3
10	IDS082N111	Elements and Principles of Design	100	3
11	PRD082N111	Sketching and Rendering	100	3
Total Credit: 23/24				

2nd Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M211	Geometrical Construction and Graphics	100	3
2	IDS082M212	Furniture Design	100	3
Interdisciplinary				
3	IKS992K201	Introduction to Indian Knowledge System – II	100	3
Ability Enhancement Course (AEC)				
4	CEN982A201	Communicative English -II	100	2
Skill Enhancement Course (SEC)				
5	IDS082S211	Computer Application -I	100	3
Value Addition Course (VAC)				
6	VAC992V2424	Vernacular Architecture of India	100	3
SWAYAM COURSE				
7	IDS082Z291	Introduction to Ancient Indian Architecture	100	3/4
Minor				
8	COD082N211	Visualization Techniques and Photography	100	3
9	FAS082N214	Fashion Styling and Photography	100	3
10	IDS082N211	Colour Theory and Application	100	3
11	PRD082N211	Product Visualization	100	3
Total Credit: 23/24				

3rd Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M311	Interior Design – I	200	4
2	IDS082M312	Interior Materials & Construction Techniques - I	200	4
Interdisciplinary				
3	IDS082I311	Bamboo & Cane crafts of North-East India	200	3
Ability Enhancement Course (AEC)				
4	CEN982A301	Communicative English-III	200	2
Skill Enhancement Course (SEC)				
5	IDS082S311	Computer Application -II	200	3

SWAYAM COURSE				
6	IDS082Z391	Interior Design by Prof. Smriti Saraswat, IIT Roorkee (8 weeks)	100	3/4
Minor				
7	COD082N311	History of Indian Visual Craft	200	4
8	FAS082N313	Fundamentals of Pattern making and Garment construction	200	4
9	IDS082N311	Art appreciation	200	4
10	PRD082N311	Form Generation	200	4
Total Credit: 23/24				

4th Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M411	Interior Design – II	200	4
2	IDS082M412	Interior Materials & Construction Techniques – II with Material Specification	200	4
3	IDS082M443	Furnishing, finishing and fittings	200	4
Ability Enhancement Course (AEC)				
4	CEN982A401	Communicative English -IV	200	2
SWAYAM COURSE				
5	IDS082Z491	Augmenting Design Thinking with Human-Computer Interaction By Prof. Sonal Atreya, IIT Roorkee (8 weeks)	200	3/4
Minor				
6	COD082N411	Brand Identity Design	200	3
7	COD082N412	Typography	200	3
8	FAS082N415	Sustainable Product Development	200	3
9	FAS082N416	Introduction to Draping	200	3
10	IDS082N411	Interior Design Representation Techniques	200	3
11	IDS082N412	Material Documentation & Specification	200	3
12	PRD082N411	Product Analysis	200	3
13	PRD082N402	Material Studies	200	3
Total Credit: 23/24				

5th Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M511	Interior Design – III	300	4
2	IDS082M512	Interior Materials & Construction Techniques – III with Material Specification	300	4

3	IDS082M501	Research Methodology	300	4
4	IDS082M543	Building Services – I	300	4
Minor				
5	COD082N511	Publication Design	300	4
6	FAS082N514	Fashion Illustrations	300	4
7	IDS082N511	Interior Landscape	300	4
8	PRD082N511	Manufacturing Processes	300	4
Total Credit: 20				

6th Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M611	Interior Design – IV	300	4
2	IDS082M612	Interior Materials & Construction Techniques – IV	300	4
3	IDS082M643	Building Services – II	300	4
4	IDS082M614	Working Drawing – I	300	4
Minor				
5	COD082N611	Copywriting	300	4
6	FAS082N615	Visual Merchandising	300	4
7	IDS082N641	Indigenous Knowledge System	300	4
8	PRD082N611	Packaging Design	300	4
Total Credit: 20				

7th Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M711	Interior Design – V	400	4
2	IDS082M703	Project Management and Estimation Costing	400	4
3	IDS082M714	Working Drawing – II	400	4
Summer Internship (SI)				
4	IDS082M721	Internship	300	4
Minor				
5	COD082N711	Advertising & Marketing	400	4
6	FAS082N515	Design Process and Portfolio	400	4
7	IDS082N715	Furnishing, finishing and fittings	400	4
8	PRD082N711	Design for Sustainability	400	4
Total Credit: 20				

NOTE : There is no multiple entry/exit in the 4 year Bachelor of Interior Design Undergraduate program and students are mandatory to do 8 weeks of Summer/Industry Internship.

8th Semester				
Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	IDS082M811	Detailed Project Report	400	4
Research Project				
2	IDS082M813	Thesis / Major Project	400	12
		OR		
A	IDS082M814	Indoor Landscape : Role of changing Indoor environment.	400	4
B	IDS082M815	Color and Materials : Role of changing Indoor environment.	400	4
C	IDS082M816	Affordable Interior Grandeur : Complete Solution of one Project	400	4
Minor				
3	COD082N812	Minor Project	400	4
4	FAS082N812	Design Portfolio & Collection Development	400	4
5	IDS082N812	Minor Project	400	4
6	PRD082N812	Minor Project	400	4
Total Credit: 20				

NOTE : BID Minor students will follow the Track of Department of Fashion Design / Communication Design/ Product Design.

Students opting for Minor tracks can select any one of the 4-track which will enable to attain a minor in the relevant domain

Minor-Track-I (Communication Design-CD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	COD082N111	Illustration Technique	100	3
2	2nd Semester	COD082N211	Visualization Techniques and Photography	100	3
3	3rd Semester	COD082N311	History of Indian Visual Craft	200	4
4	4th Semester	COD082N411	Brand Identity Design	200	3
		COD082N412	Typography	200	3
5	5th Semester	COD082N511	Publication Design	300	4
6	6th Semester	COD082N611	Copywriting	300	4
7	7th Semester	COD082N711	Advertising & Marketing	400	4

8	8th Semester	COD082N812	Minor Project	400	4
Total Credit: 32					

Minor-Track-II (Fashion Design-FD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	FAS082N114	Fashion Design and Concept	100	3
2	2nd Semester	FAS082N214	Fashion Styling and Photography	100	3
3	3rd Semester	FAS082N313	Fundamentals of Pattern making and Garment construction	200	4
4	4th Semester	FAS082N415	Sustainable Product Development	200	3
		FAS082N416	Introduction to Draping	200	3
5	5th Semester	FAS082N514	Fashion Illustrations	300	4
6	6th Semester	FAS082N615	Visual Merchandising	300	4
6	7th Semester	FAS082N515	Design Process and Portfolio	400	4
7	8th Semester	FAS082N812	Design Portfolio & Collection Development	400	4
Total Credit: 32					

Minor Track-III (Interior Design-ID)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	IDS082N111	Elements and Principles of Design	100	3
2	2nd Semester	IDS082N211	Colour Theory and Application	100	3
3	3rd Semester	IDS082N311	Art appreciation	200	4
4	4th Semester	IDS082N411	Interior Design Representation Techniques	200	3
		IDS082N412	Material Documentation & Specification	200	3
5	5th Semester	IDS082N511	Interior Landscape	300	4
6	6th Semester	IDS082N641	Indigenous Knowledge System	300	4
6	7th Semester	IDS082N715	Furnishing, finishing and fittings	400	4
7	8th Semester	IDS082N812	Minor Project	400	4
Total Credit: 32					

Minor Track-IV (Product Design-PD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	PRD082N111	Sketching and Rendering	100	3
2	2nd Semester	PRD082N211	Product Visualization	100	3
3	3rd Semester	PRD082N311	Form Generation	200	4
4	4th Semester	PRD082N411	Product Analysis	200	3
		PRD082N402	Material Studies	200	3
5	5th Semester	PRD082N511	Manufacturing Processes	300	4
6	6th Semester	PRD082N611	Packaging Design	300	4
7	7th Semester	PRD082N711	Design for Sustainability	400	4
8	8th Semester	PRD082N812	Minor Project	400	4
Total Credit: 32					

1st Semester		
Paper 1 Major	**Design Thinking** **L-T-P-C : 1-0-2-3 Credits : 3** **Scheme of Evaluation : Studio**	**Subject Code: IDS082M111** **Level : 100**

Course Objective: The objective of the course **Design Thinking (IDS082M111)** is to develop cognitive, strategic and practical thinking and ideation processes by which design concepts are developed.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Remember the use of abducted and productive reasoning	BT1
CO 2	Understand the principles, theories and models of design thinking	BT2
CO 3	Apply and adopt solution focus strategies.	BT3
CO 4	Analyze and resolve design problems in creative ways.	BT4

Prerequisite:

- No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Design Thinking** What is Design? What is Design thinking? Design Thinking models	6	15
II	**Basic Elements of Design** Simplicity, Unity, Proportion, Emphasis, Rhythm, Balance Glossary of technical words, natural structures and intuitive understanding their behavior	6	15
III	**Thinking Techniques** Principles in generative, convergent, lateral, interactive, check lists, analysis and synthesis simulation, and implementations of intentions.	5	15
IV	**Design tools** Mind mapping, Empathy mapping, Brain storming, Positive techniques for creativity, Creative pause, Focus, Challenge, alternatives, concepts, sensitizing techniques, group or individual techniques. Simple design exercises.	5	15
	TOTAL	**82**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			Poster making, Group Work, Debates, Discussions and Presentations

Text Books:

1. Nigel Cross (2011); Design Thinking: Understanding How Designers Think and Work, 1st Ed.

Reference Books:

1. Scott,R.G (2009); Design Fundamentals; 2nd Ed.; R.E. Kreiger; New York
2. Itten, Johannes (1970) Elements of colours; 1st Ed.; Chapman & Hall

Reading Materials:

1. Brown. T. (2008) Design Thinking Thoughts: Definitions of Design Thinking retrieved from https://designthinking.ideo.com/?p=49
2. Kolko, J. (2015, Sep) The Evolution of Design Thinking. Harvard Business Review
3. Liedtka, J. et al (2013) Solving Problems with Design Thinking: Ten Stories of What Works. Columbia University Press
4. Osborn, A. (1953/2001) Applied Imagination: Principles and Procedures of Creative Problem Solving. Creative Education Foundation Press.
5. Liedtka, J. (2018, Sept-Oct) Why Design Thinking Works, Harvard Business Review

1st Semester		
Paper 2 Major	**Basic Design** **L-T-P-C : 1-0-2-3 Credits : 3** **Scheme of Evaluation : Studio**	**Subject Code: IDS082M211** **Level : 100**

Course Objective:

The objective of the course **Basic Design (IDS082M211)** is to develop a design sense and apply the basic design acumen, anthropometric observations in the visualization/ composition of spaces.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define design principles, basic human factors, anthropometrics, and ergonomics.	BT1
CO 2	Demonstrate an ability to understand the importance of human factors, human proportions in designing functional spaces.	BT2
CO 3	Apply principles of design, ergonomic aspects, and anthropometrical data of various activities for design development.	BT3
CO 4	Examine interior work systems with respect to design elements, principles, and human factors.	BT4

Prerequisite:

Basic concepts of human scale and proportion along with technical drafting.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Studio)	Design Principles • Lines: properties and nature of lines along with usage • Shape & Forms: organic/ geometric shapes and understanding their role w.r.t space (positive/negative) • Textures: tactile texture, visual texture and how they are implied • Colour: Hue-Saturation, Value, Intensity, Nature of Colour, and implications, colour schemes • Principles of Design: Balance (symmetrical/asymmetrical), Contrast (anomaly, emphasis, subordination, Rhythm/ repetition, Proportion/scale, Harmony and unity, Variety, Radiation	6	15
II (Theory)	Introduction to Anthropometry, Ergonomics, associated Terminologies, and its Relevance Learning about human scale/ dimensions, anthropometry, understanding human proportions for various activities and its spatial aspects, human aid to lifestyle, environmental factors influencing human performance, physiology (work physiology) and stress.	6	15

III (Theory)	Human Physical Dimension, Human Body Structure and Function • Static and dynamic anthropometry, stand posture- erect, anthropometry landmark: sitting postures, anthropometry: squatting and cross legged postures. • Posture and activity relation, posture and body supportive devices, chair characteristics, vertical surface, horizontal surface, movement, work counter (standing & seating).	5	15
IV (Studio)	Application through Design: • Application of principles of design in: Room arrangement, Interior display and Furniture, Carpets. • Application of Ergonomics in Residential spaces.	5	15
	TOTAL	**82**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
3	22 hours	60 hours	8 hours
			Measured Drawing Group work/ presentation Ergonomics Model & Report Making

Text Book:

1. Elizabeth. D. Hutchinson; *Dimensions of Human Behaviour, Person and Environment*, Sage Publications, 2007.
2. Francis D.K. Ching, *Form, Space and Order;* John Wiley and Son, 2014
3. Charlotte & Peter Fiell; *Modern Furniture Classics Since 1945,* Thames & Hudson
4. Alan Barnard & Jonathan Spence; *Encyclopedia of Social and Cultural Anthropology*, Routledge Edition 1, 2002

Reference Books:

1. De Chiara Joseph & Callender John; *Time Saver Standards for Architectural Types & Interior Design & Space Planning*; McGraw Hill Book Co.
2. *Time Saver Standards for Building Types*; De Chiara, Joseph and Crosbie, Michael J.*;* 2nd Ed. 2011; Tata McGraw Hill; New Delhi
3. Neufert, Peter; *Neufert's Architects' Data;* 4th Ed.; 2012; John Wiley; New Delhi
4. Gerald F. Brommer; *Elements and Principles of Design: Student Guide with Activities*, 2004

Study Material:

1. Linda L. Nussbaumer*; Human Factors in the Built Environment;* Bloomsbury Academic; January 2018, Edition 2
2. Grandjean E, *Fitting the Task to the Man*; Taylor & Francis, London, 1988

1st Semester		
Paper 3 Minor	**Elements and Principles of Design** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code: IDS082N111** **Level : 100**

Course Objective:

The objective of the course **Elements and Principles of Design (IDS082N111)** is to develop an understanding of elements of design in abstract and application of Principles of Design and Appraise the design of form to help create an aesthetically pleasing & creative design.

Course Outcomes:

After successful completion of the course, student will be able to:

CO Level	Course Outcome	Blooms Taxonomy Level
CO1	Understanding and applying design vocabulary such as point, line, shape, color, texture, area, mass, volume etc.	**BT1**
CO2	Understanding and applying design principles such as ratio, proportion, scale, balance, harmony, unity, variety, rhythm, emphasis.	**BT2**
CO3	Understanding the value of coherently incorporating; drawing, sketching and model making, into an acceptable professional standard of communication, using materials and techniques to coordinate the design process.	**BT2**
CO4	Follow the design process to create a creative and stimulating form/space, using the basic strategic knowledge and tactics of; structure, line, form, space, lighting and shape.	**BT3, BT4**

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods	Hours
		L	P
I	**Design Elements** Lines: properties and nature of lines along with usage SHAPE AND FORMS: organic/ geometric shapes and understanding their role w.r.t space (positive/negative)	2	16
II	**Texture and colour** Textures: tactile texture, visual texture and how they are implied Colour: HUE-SATURATION, VALUE, INTENSITY nature of Colour, implications	2	16
III	**Design principles** Balance (symmetrical/asymmetrical) Contrast (anomaly, emphasis, subordination) Rhythm/ repetition Proportion/scale Harmony and unity. Variety	2	16

	Radiation		
IV	**Basics of Graphic Design** Graphic Design, logo design, tile border design, Wall mural	2	16
TOTAL		72	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			True Scale representations, Measure drawing, Exposure to different mediums (Charcoal, Acryclic etc.)

Text Books:

1. *Principles of two dimensional design*; Wong, Wucius*;* 1st Ed.; 1972; John Wiley; New York.

Reference Books:

1. Wong, Wucius; *Principles of three dimensional design*; 1st Ed.; 1977; Van Nostrand; New York
2. Christopher Alexander. *A Pattern language: towns, buildings, construction*. Oxford University Press, 1978.

Semester – I		
Paper 4/Subject Name: Introduction to Indian Knowledge System – I **Subject Code: IKS992K101** **L-T-P-C – 2-1-0-3** **Credit Units: 3** **Course Level: 100** **Scheme of Evaluation: Summative Assessment - 70% + Formative Assessment -30%**		
Credit Distribution (NCH)		
Lecture/Tutorial (Hours)	**Practical (Hours)**	**Experiential Learning (Hours)**
60	0	30

Course objectives:

This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes:

On completion of this course students will be able to :

CO	Contents	BT Level
CO_1	Recall the rich heritage of Indian knowledge systems	BT level 1
CO_2	Describe the contribution of Indian knowledge systems to the world	BT level 2
CO_3	Demonstrate knowledge of sociocultural and ethnolinguistic diversity that constitutes the soul of Bharatvarsha	BT level 2
CO_4	Apply traditional knowledge and techniques in day-to-day life	BT level 3
CO_5	Distinguish knowledge traditions that originated in the Indian subcontinent	BT level 3

Module	Course Contents	Periods
I	**Introduction to Indian Knowledge Systems (IKS):** About Indian Knowledge System; Definition of Indigenous/ Traditional Knowledge; Scope, and Importance of Traditional Knowledge. **Ancient India- Bharat Varsha:** People of Ancient Bharat Varsha; Our great natural heritage: The great Himalayas and the rivers; The civilizations of the Sindhu-Ganga valley, and the Brahmaputra valley; Our coastal plains; Our Nature: Forests and Minerals; Ancient Indian Traditional Knowledge and Wisdom about nature and climate.	15

Module	Course Contents	Periods
II	**Indian Heritage of Knowledge:** Ancient Indian Knowledge: The *Vedas* and its components-the *Vedangas* Ancient Indian books and treaties: The *Sastras*.; The Great Indian Epics: The Ramayana and The Mahabharata Epics and religious treaties of ancient Assam: Introduction to Madhav Kandali's *Ramayan* and Srimanta Sankardev's *Dasam Skandha Bhagavat* of the Puranas; Ancient Traditional Knowledge-The *Agamas ;* The ancient Buddhist knowledge: *Tripitaka: Vinaya, Sutta* and *Abhidhamma Pitaka* **Languages and language studies in India**: What is linguistics?; Script and Language; Alphabet of the Indian; languages *Varnamala:* Origin, Evolution, and phonetic features; Languages of India; Important texts of Indian languages: Skills *Siksha,* Expression/Prounciation-*Nirukta,* Grammer-*Vyakarana,* Poetic rhythm-*Chandas;*Paninian Grammar: A Brief Introduction **Introduction to Fine Arts and Performing Arts of India:** Ancient Indian classical music and dance forms: The Science of Dramas-*Natyasastra* and the Science of Music-*Gandharva-Veda;* Aesthetics in Indian Art and Culture; Folk music and traditional dance forms of the Northeast.	15
III	**Indian Science & Technology:** Ancient India's contribution to Mathematics - Number System. Algebra and Arithmetic, Geometry and Trigonometry; Origin of Decimal system in India; nomenclature of numbers in the Vedas. Zero and Infinity. Sulba-sutras. Contribution of Brahmagupta and Sridhar Acharya to Mathematics. Important texts of Indian mathematics. **Indian Astronomy:** Planetary System. Motion of the Planets; Velocity of Light; Eclipse. Astronomy. Navagrahas. Important works in Indian Astronomy. Aryabhata and Nilakantha: Contribution to Astronomical Studies **Indian Metal Works:** Mining Techniques. Types of Metals. Tools & Techniques for Metal Smelting with examples. Metalworks in pre-modern India: Special reference to NE India.	15
IV	**Contribution of Ancient India to Health Sciences:** Traditional Indigenous systems of medicines in India: - *Ayurveda* and *Yoga;* Elements of *Ayurveda: Gunas* and *Doshas*, *Pancha Mahabhuta* and *Sapta-dhatu;* Concept of disease in Ayurveda; Ayurvedic lifestyle practices: *Dinacharya* and *Ritucharya;* Important Ayurvedic Texts; Hospitals in Ancient India; *Ayurveda*: Gift of India to the modern world.	15
Total		**60**

Textbooks Books:

1. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
2. Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

1. Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.
2. D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.
3. Astāngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
4. Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5. J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

The experiential learning sessions may include:
• Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems. • Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge. • Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System. • Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of Indian knowledge systems. • Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.

1st Semester		
Paper 5 AEC	**CEN I: Introduction to Effective Communication** **L-T-P-C : 1-0-0-1 Credits : 1** **Scheme of Evaluation : Theory+Practical**	**Subject Code: CEN982A101** **Level : 100**

Course Objective: To understand the four major aspects of communication by closely examining the processes and outlining the most effective ways to communicate with interactive activities.

Course Outcomes: On successful completion of the course the students will be able to

CO Level	Course Outcome	Blooms Taxonomy Level
CO 1	**List** the elements and processes that make for successful communication and **recognise** everyday activities that deserve closer attention in order to improve communication skills	**BT 1**
CO 2	**Contrast** situations that create barriers to effective communication and **relate** them to methods that are consciously devised to overcome such hindrance	**BT 2**
CO 3	**Apply** language, gestures, and para-language effectively to avoid miscommunication and **articulate** one's thoughts and build arguments more effectively	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	Introduction to Effective Communication • Listening Skills o The Art of Listening o Factors that affect Listening o Characteristics of Effective Listening o Guidelines for improving Listening skills	**5**
II	• Speaking Skills o The Art of Speaking o Styles of Speaking o Guidelines for improving Speaking skills o Oral Communication: importance, guidelines, and barriers	**5**

III	• Reading Skills o The Art of Reading o Styles of Reading: skimming, surveying, scanning o Guidelines for developing Reading skills	**5**
IV	• Writing Skills o The Art of Writing o Purpose and Clarity in Writing o Principles of Effective Writing	**5**

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
15 hours	-	10 hours - Movie/ Documentary /Podcasts screening - Peer teaching

Text Books:

1. Rizvi, M. Ashraf. (2017). *Effective Technical Communication*. McGraw-Hill.
2. Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). *Business Communication.* Pearson.
3. Raman, Meenakshi and Sharma, Sangeeta. (2011). *Technical Communication: Principles and Practice* (2nd Edition): Oxford University Press.

1st Semester		
Paper 6 SEC	**Joinery and Surface Aesthetics** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practicum/Jury**	**Subject Code: IDS082S111** **Level : 100**

Course Objective: The objective of the course **Carpentry Workshop (IDS082S111)** is to give an introduction to the basic principles governing structural systems.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define basic knowledge of working with structural materials	BT1
CO 2	Interpret the techniques of planning, chiselling, pasting, fixing and welding to learn the use of hand tools.	BT2
CO 3	Make use of practical usage of color and its application.	BT3
CO 4	Discover skill in creating designs and making art objects	BT4

Prerequisite:

- Basic concepts of art and scales.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I (Practical)	**Installation (Part I) Carpentry (Timber):** Introduction t the carpentry tools, processes, joints and wood working machines. Preparation of various carpentry joints, fixing of plywood. Blackboards, commercial boards etc. and their application of furniture. **Installation (Part II)** Handling materials like POP. Gypsum. Aluminium etc. Understanding nailing. Screwing, riveting and their various conditions and types of applications.	6	15
II (Practical)	**Fabrication Installation** Introduction to welding equipment, processes and its applications. Hands on experience to come up with a product or installation. **Masonry Installation** Installations with bricks and brick bats. Making of geometrical forms on the ground.	6	15
III (Practical)	**Introduction to Color** Understanding color Properties of color What is color	5	15

	Responses to color Color terms **Color principles** Black and white, Neutral,Chromatic, Value, Hue. Color solid, complimentary hues. Color harmony & simultaneous contrast		
IV (Practical)	**Color design** Design with Value, Chroma and Hue gradations. Design with hue mixtures, complimentary hues & unrelated hues. Developing a color scheme. Color expression: color guide **Project Work** Problem Solving (Residential / Commercial)/application of theory.	5	15
	TOTAL	**82**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			Hands on assessment, Tool handling, Field Visit and Working with craftsmen.

Text Books:

1. Mackay, W.B. (2005) Building construction. 1st Ed. Donheed, London.

Reference Books:

1. Chudley, R. (1987) Construction technology. 2nd Ed. ELBS; Harlow
2. Simmons H. L. (2007) Olin's Construction Principles, Materials and Methods, John Wiley and Sons.
3. *Principles of Color Design:* Wong Wucius.; 2nd edition, John Wiley & Sons, Inc.

Reading Materials:

1. Branco, J. M., & Thierry, D. (2015). Analysis and Strenghtening of Carpentry Joints. Construction and Building Materials , 34-47.
2. *Color Matching Handbook,* Thunder Bay Press, California.
3. Michael E. Doyle; *Color Drawing*; 3rd Edition, John Wiley & Sons, Inc.

1st Semester		
Paper 7 VAC	**Indian Architectural Heritage** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code: VAC992V1413** **Level : 100**

Course Objective:

The objective of the course **Indian Architectural Heritage (VAC992V1413)** is to understand the evolution of Buddhist, Temple and Islamic Architecture in India in their various stylistic modes characterized by technology, ornamentation and planning practices. To provide an understanding and appreciation of Contemporary Architecture through the various Art and Architectural movement, theory, works of master architects in India.

Course Outcomes:

After successful completion of the course, student will be able to		
S.No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	To define the basic knowledge of Buddhist, Temple, Islamic and Contemporary Architecture of India	BT1
CO2	Outline the origins and evolution of traditions and regional styles of various temples, monuments and structures in India	BT2
CO3	Develop an understanding of the historical and social context of various built forms of different eras/periods.	BT3

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods	Hours
		L	P
I	**Buddhist Architecture** Early Buddhist Cultures (Design norms, standards, prescription and style) and philosophies Viharas Great Stupa of Sanchi, etc., Hinayana & Mahayana Phase	2	8
II	**Temple Architecture** Evolution of Hindu temple –Indo Aryan and Dravidian Beginning of Dravidian architecture Pallavas, Rathas at Mamallapuram, Shore Temple Kailashnatha and Vaikuntaperunal Temple at Kancheepuram	2	8
III	**Islamic Architecture** Monumental buildings in the regime of various Mughal rulers (Akbar, Humayun, Shah Jahan) A Study of prominent monuments of the era (Fort of Agra,	2	8

	Gwalior Palace, Taj Mahal etc)		
IV	**Miscellaneous** Influence of Global Architects - Le Corbusier, Louis I Kahn Post-independence architecture New ideologies – continuity – Ganesh Bhikaji Deolalikar Revivalism – BR Manickam Modernism and Critical regionalism – BV Doshi, Raj Rewal, Charles Correa, Laurie Baker	2	8
TOTAL		40	

Notional Credit Hours for the course: 16 x 3 = 48

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	8 hours	32 hours	8 hours
			Measured Drawing Group work/ presentation Ergonomics Model & Report Making

Text Books:

1. History of Architecture in India by Tadgell Christopher
2. Indian Architecture: Buddhist and Hindu Period by Percy Brown

Reference Books :

1. A Concise History of Modern Architecture in India by Jon Lang
2. Masterpieces of Traditional Indian Architecture by Satish Grover
3. Concepts of Space in Traditional Indian Architecture by Yatin Pandya

2nd Semester		
Paper 1 Major	**Geometrical Construction and Graphics** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Studio/Jury**	**Subject Code: IDS082M211** **Level : 100**

Course Objective: The objective of the course **Geometrical Construction and Graphics (IDS082M211)** is to orient and familiarize the students towards the principles of Design thinking, Visualization and representation with drawing materials and equipment.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Choose rendering techniques for representing 2D and 3D drawings.	BT1
CO 2	Demonstrate sciography, Human figures and accessories in 2D and 3D drawings.	BT2
CO 3	Apply Learning of measure drawing with basic knowledge and understanding of Isometric, Oblique, Axonometric projections	BT3
CO 4	Analyze 3D visualization based on perspective and sciography.	BT4

Prerequisite:

- Basic concepts of Design Thinking and Drafting.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Studio)	**Basics of Geometrical Construction-** Construction of basic shapes: lines, angles, triangles, squares, rectangles, circles. Understanding properties of shapes.	6	15
II (Studio)	**Isometric & Axonometric Projections** Preparation of views of basic figures	6	15
III (Studio)	**One Point Perspective** Picture plane, Vanishing Point, Eye level, Station Point. Draw Objects **Two Point Perspective** Picture plane, Vanishing Point, Eye level, Station Point. Draw Objects	5	15
IV (Studio)	**Sciography-1 and 2** Learning sociography of simple objects, Sociography in	5	15

	Perspective		
	TOTAL	**82**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			True Scale representations, Measure drawing, Exposure to different mediums (Charcoal, Acryclic etc.)

Text Books:

1. Bhatt, N.D and Panchal, V. M. (2008). Engineering drawing, 1st Ed. Charotar Publishing House, Gujrat.

Reference Books:

1. John Montague, Willey, (1985), Basic Perspective Drawing, A Visual Approach, 6th Edition, John Willey and sons, Inc.
2. Francis D.K. Ching, (1998), Design Drawing, John Willey and sons, Inc.
3. Mulik, Shankar, (1994) A Text Book of perspective & Sciography, Allied Publishers Ltd., Mumbai.
4. Sherkey W, Morgan. (1950). Architectural Drawing: Perspective, Light and Shadow, Rendering, Mc Graw Hill.
5. Arthur L. Guptill, Watson. (1997). Rendering in Pen and Ink, Guptill Publications, New York.

Reading Materials:

Blankenbehler, B. (2015, October 19). *Shadow In Perspective Drawing- Art Technique*. Retrieved December 14, 2022, from Architecture Revived: https://architecturerevived.com/shadow-in-perspective-drawing-art-technique/

Branco, J. M., & Thierry, D. (2015). Analysis and Strenghtening of Carpentry Joints. *Construction and Building Materials*, 34-47.

C.F.Reinhart. (2002). Effects of interior design on the daylight availability in open plan. *Conference Proceedings of the ACEEE Summer Study on Energy Efficient Buildings.* California: NRC Publications Archive.

Carlson, C. (2022, December 12). *Dezeen's top 10 home interiors of 2022*. Retrieved December 14, 2022, from dezeen: https://www.dezeen.com/2022/12/12/top-home-interiors-2022-review/

Crannell, A. (2010). Perspective drawings of reflective spheres. *Journal of Mathematics and the Arts*, 71-85.

Gou, Z., Lau, S.-Y. S., & Shen, J. (2012). Indoor Environmental Satisfaction in Two LEED Offices and its Implications in Green Interior Design. *Indoor and Built Environment*, 503-514.

Nabil, S., & Kirk, D. (2021). Decoraction: a Catalogue for Interactive Home Decor of the Nearest-Future. *Proceedings of the Fifteenth International Conference Tangible, Embedded, and Embodied Interaction* (pp. 1-13). Salzburg: Association for Computing Machinery, New York.

Olkun, S. (2003). Making Connections: Improving Spatial Abilities with Engineering Drawing Activities. *International Journal of Mathematics Teaching and Learning*.

Werner, C. M. (1987). Home Interiors: A Time and Place for Interpersonal Relationships. *Environment and Behaviour*.

Wright, F. L., Samona, G., & Mayor, A. H. (1960). Drawings for a Living Architecture. *Journal of the Society of Architectural Historians*, 129-131.

SYLLABUS (2nd SEMESTER)

Paper /Subject Name: Furniture Design		**Subject Code: IDS082M212**
L-T-P-C: 0-0-6-3	**Credit Units: 3**	**Scheme of Evaluation: S**

Objective:

The objective of the course **Furniture Design I (IDS132D311)** is to design furniture for different rooms/spaces of the residence designed in 2nd Semester.

Course Outcome:

On completion of this course, students will be able to

CO1: How to design and create furniture pieces for different residential spaces understanding its relation to human movements.

CO2: Demonstrate the multiple use of furniture for different residential spaces keeping the constraints of available space in consideration.

CO3: Develop principles that will help one to judge the design of furniture for different residential spaces in relation to their functionality and aesthetics.

CO4: Analyze the use of contextually appropriate materials, colour scheme, texture, joinery details to care & maintain the furniture for different residential spaces with various finishes.

Prerequisite:

- **Basic concepts of design thinking and technical drafting.**

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods
I	**Introduction and History of Furniture Design** Definitions, Styles, Significance, Comparative analysis of different styles. Design Process: User profiling, Anthropometric study, Design development	**24**
II	**Living and Dining Room** Design furniture pieces for living and dining spaces	**24**
III	**Bedroom – Kids and Master** Design furniture for bedroom spaces	**24**
IV	**Thematic residential spaces** Barrier free, Transition spaces, Home office space, Hobby areas etc	**24**
	TOTAL	**96**

Text Book:

5. Shah Kavita ,Desai Nidhi, Patel Dr. Sarjoo ,Trivedi Khyati, Residential Space Designing, Authorspress, 2020.
6. Charlotte & Peter Fiell, *Modern Furniture Classics Since 1945,* Thames & Hudson
7. Darby Tom, *Making Fine Furniture*; Guild of Master Craftsman Publications.

Reference Books:

5. De Chiara Joseph &Callender John, *Time Saver Standards for Architectural Types & Interior Design & Space Planning*; McGraw Hill Book Co.
6. Gilliatt Mary, *The Decorating Book*; Michael Joseph

Reading Material:

3. Grandjean E, *Fitting the Task to the Man*; Taylor & Francis, London, 1988
4. Lawson Stuart, *FURNITURE DESIGN: An Introduction to Development, Materials, Manufacturing,* Laurence King Publishing, 2013.

2nd Semester		
Paper 3 Minor	**Colour Theory and Application** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code: IDS082N211** **Level : 100**

Course Objective:

The objective of the course **Colour Theory and Application** (**IDS082N211**) introduce students to the concept of application of colours and to understand the judicious use of colours in interior design.

Course Outcomes:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Understand the evolution of color theory and learn color terminology/vocabulary and how to communicate/define color. Develop knowledge regarding the sensitive to the psychological effects of color,	BT1
CO2	Explore various color systems and the ways they are applied today and similarities and differences among the relationship and interaction of color	BT2
CO3	Create color harmony (monochromatic, analogous, complementary, split complementary, and triadic) in a color composition.	BT3
CO4	Observe various occurrences and uses of color in nature, art, architecture, fashion/textiles, film, interior design, graphic design and photography.	BT 4

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods
I	**Introduction to Color** Understanding color Properties of color What is color Responses to color Color terms	12
II	**Color principles** Black and white, Neutral,Chromatic, Value, Hue. Color solid, complimentary hues. Color harmony & simultaneous contrast	16
III	**Color design** Design with Value, Chroma and Hue gradations. Design with hue mixtures, complimentary hues & unrelated hues.	24

	Developing a color scheme. Color expression:color guide	
IV	**Project work** Problem Solving (Residential / Commercial)/application of theory.	12
TOTAL		64

Text Books:

3. *Principles of Color Design:* Wong Wucius.; 2nd edition, John Wiley & Sons, Inc.

Reference Books:

1. *Color Matching Handbook,* Thunder Bay Press, California.
2. Michael E. Doyle; *Color Drawing*; 3rd Edition, John Wiley & Sons, Inc.

Semester – II		
Paper 4/Subject Name: Introduction to Indian Knowledge System - II		
Subject Code: IKS992K201		**L-T-P-C – 2-1-0-3**
Credit Units: 3		**Course Level: 100**
Scheme of Evaluation: Summative Assessment - 70% + Formative Assessment -30%		
Credit Distribution (NCH)		
Lecture/Tutorial (Hours)	**Practical (Hours)**	**Experiential Learning (Hours)**
60	0	30

Course objectives:

This Foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG program. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Course Outcomes:

On completion of this course, students will be expected to –

CO	Contents	BT Level
CO_1	Recall traditional Indian knowledge traditions constituting Indian culture	BT level 1
CO_2	Summarize differences between classical literature in Sanskrit and other Indian languages	BT level 2
CO_3	Compare knowledge traditions originating in NE India	BT level 2
CO_4	Appreciate the contribution of Indian Knowledge Systems to the world	BT level 3

Module	Course Contents	Periods
I	**Indian Classical Literature** Indian Classical Literature: A Brief Introduction; Ancient Indian Spritual Poetics-*Kavya*: Contribution of Kalidasa **Diversity and Indian Culture**: Diversity and Indian Culture; Indigenous Faith and Religion; Preservation of culture and indigenous knowledge **The Purpose of Knowledge:** Understanding Self-Awareness and Spirituality; Indian concept and purpose of Knowledge and Education; Understanding Spirituality and Materialism: *Para* and *Apara Vidya*	15

Module	Course Contents	Periods
II	**Methodology of Indian Knowledge System**: *Shruti* and *Smriti* traditions; Intoduction to *Shastras;* Manuscriptology: The art and science of documenting knowledge; Repositories of ancient manuscripts with special reference to the Norteast India. **Indian Architecture and Town Planning**: Introduction ancient Indian architecture; *Sthapatya-Veda*: An Introduction; Indigenous tools & techniques for town planning & Temple Architecture. Lothal, Mohan Jo Daro; Temple Art: Lepakshi Temple, Jagannath Puri Temple, Konark Sun Temple; Vernacular architecture of Assam: Special reference to Brahmaputra Valley	15
III	**Indian Agriculture**: Agriculture: Significance in Human Civilization; Sustainable Agriculture; Historical significance of agriculture and sustainable farming in India;Step Cultivation of India: Special reference to Northeast India; Wet rice cultivation of Assam. **Indian Textiles**: What is Textile?; Tradition of cotton and silk textiles in India; The historical contribution of textile and weaving to the Indian economy; Varieties of textiles and dyes developed in different regions of India with special reference to Northeast India	15
IV	**Indian Polity and Economy**: Understanding Kingdom and Chiefdom; Role of a king; The Indian idea of a well-organized polity and flourishing economy; The *Chakravarti* System: Administrative System of Ancient Bharatvarsha; Village administrative system: Northeast India; *Arthashastra*: Brief synopsis **The outreach of Indian Knowledge System across Geographical Boundaries:** Indian Languages; Scripts; Linguistics; Ayurveda; Yoga and Meditation; Textile; Decimal value place system-based arithmetic, Algebra and Astronomy	15
Total		**60**

Textbooks Books:

1. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022), *Introduction to Indian Knowledge System: Concepts and Applications.* PHI Learning Private Ltd.
2. Mukul Chandra Bora*, Foundations of Bharatiya Knowledge System.* Khanna Book Publishing

Reference Books:

1. Baladev Upadhyaya, *Samskrta Śāstrom ka Itihās*, Chowkhambha, Varanasi, 2010.
2. D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., *A Concise History of Science in India*, 2nd Ed., Universities Press, Hyderabad, 2010.

3. Astāngahrdaya, Vol. I, *Sūtrasthāna and Śarīrasthāna*, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
4. Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5. J. K. Bajaj and M. D. Srinivas, *Indian Economy, and Polity in Eighteenth-century Chengalpattu*, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.

The experiential learning sessions may include:

- Field Visits: Organizing visits to historical sites, museums, traditional craft centers, and other places relevant to Indian knowledge systems.
- Interactive Sessions: Engaging students in discussions with experts and practitioners in various fields of Indian knowledge systems to gain insights and practical knowledge.
- Online Lecture Series: Providing the students with online lectures by distinguished experts in the field of the Indian Knowledge System.
- Hands-on Activities: Providing opportunities for students to participate in activities related to traditional arts, crafts, music, dance, agriculture, etc., to understand the practical aspects of Indian knowledge systems.
- Practical Demonstrations: Conducting workshops or sessions to demonstrate traditional practices, such as yoga, Ayurveda, Vastu Shastra, etc., for the students.

2nd Semester		
Paper 5 **AEC**	**CEN II: Approaches to Verbal and Non-Verbal Communication** **L-T-P-C : 1-0-0-1 Credits : 1** **Scheme of Evaluation : Theory+Practical**	**Subject Code:** CEN982A201 **Level : 100**

Course Objectives :

To introduce the students to the various forms of technical communication and enhance their knowledge in the application of both verbal and non-verbal skills in communicative processes.

Course Outcomes :

On successful completion of the course the students will be able to:		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the different types of technical communication, their characteristics, their advantages and disadvantages.	**BT 1**
CO 2	**Explain** the barriers to communication and ways to overcome them.	**BT 2**
CO 3	**Identify** the means to enhance conversation skills.	**BT 3**
CO 4	**Determine** the different types of non-verbal communication and their significance.	**BT 4**

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods**
I	**Technology Enabled Communication** Communicating about technical or specialized topics, Different forms of technology-enabled communication tools used in organizations Telephone, Teleconferencing, Fax, Email, Instant messaging, Blog, Podcast, Videos, videoconferencing, social media	**4**
II	**Communication Barriers** Types of barriers: Semantic, Psychological, Organisational, Cultural, Physical, Physiological, Methods to overcome barriers to communication.	**4**
III	**Conversation skills/Verbal Communication** Conversation – Types of Conversation, Strategies for Effectiveness, Conversation Practice, Persuasive Functions in Conversation, Telephonic Conversation and Etiquette Dialogue Writing, Conversation Control.	**4**
IV	**Non-verbal Communication** Body language- Personal Appearance, Postures, Gestures, Eye Contact, Facial expressions Paralinguistic Features-Rate, Pause, Volume, Pitch/Intonation/ Voice/Modulation, Proxemics, Haptics, Artifacts, Chronemics,	**4**
	Total	**16**

Text Books:

1. Rizvi, M. Ashraf. (2017). *Effective Technical Communication*. McGraw-Hill.
2. Chaturvedi, P. D. and Chaturvedi, Mukesh. (2014). *Business Communication.* Pearson.
3. Raman, Meenakshi and Sharma, Sangeeta. (2011). *Technical Communication: Principles and Practice* (2nd Edition): Oxford University Press.

2nd Semester		
Paper 7 SEC	**Computer Application I** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation: Practicum/Jury**	**Subject Code:** IDS082S211 **Level : 100**

Course Objective: The objective of the course **Computer Application – I (IDS082S211)** is to familiarize students with basic usages of computer applications in making drawings and orient students to the digital tools in the field of Interior Design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Show the applications of computers in interior design.	BT1
CO 2	Illustrate the usage of applications of computers in interior design.	BT2
CO 3	Experiment with where to use software interior design.	BT3
CO 4	Inference the methods of digital representation in interior design.	BT4

Prerequisite:

- Basic knowledge of computer.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	Introduction to basics of AutoCAD: 1. Familiarization with AutoCAD interface. 2. Setting up drawing: standard units. 3. Introduction to drawing properties tab. 4. Setting up layers. 5. Create and Save AutoCAD drawing files 6. Define Positions using the Basic Entry methods.	6	15
II	Learning to creating basic drawings in AutoCAD. Introduction to use of various commands used for drawing creation: LINE. PLINE, REC, CIRCLE, ARC, OSNAP, etc. Introduction to use of various commands used for drawing modification: MOVE, COPY, TRIM, DIVIDE, OFFSET, CHAMFER, EXPLODE, ERASE, etc. Creation of a composition using basic geometrical shapes in AutoCAD.	6	15

III	Drawing of plan, section and elevation of a small residence in AutoCAD. i. Drawing of walls using LINE, PLINE, SPLINE, etc. ii. Creation of door and window using TRIM, CIRCLE, etc. iii. Creation of floor tiles using HATCH. iv. Creation of labels using MTEXT. v. Adding dimension using DIMENSION, DIMSTYLE. vi. Drawing and adding furniture using BLOCK, IMPORT.	5	15
IV	Learning to print (plot) a drawing in AutoCAD: Page layout for printing and Viewport. Introduction to page setup manager. Setting required paper size and printer (plotter) for printing (A1, A2, A3). Setting Viewport margins and assigning plot area. Setting Plot scale, plot style table and paper orientation.	5	15
	TOTAL	**82**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			Group Work Presentation/ Report Making Office Visits

Text Books:

1. *Up and Running with AutoCAD 2015: 2D & 3D Drawing and Modelling*, Gindis, E;*1st Ed.; 2014;* Elsevier; London.

Reference Books:

1. Seidler, D. R.; *Digital Drawing for Designers: A Visual Guide to AutoCAD 2012;* 1st Ed.; 2007; Fairchild Publications;London.
2. Moss, Elise; *Autodesk AutoCAD Architecture 2016 Fundamentals;* 1st Ed.; 2011; SDC Publications; London.
3. Omura, George & Benton, Brian C; *Mastering AutoCAD 2016 and AutoCAD LT 2016;* 1st Ed.; 2017; John Wiley &Sons; London.

2nd Semester		
Paper 8 VAC	**Vernacular Architecture of India** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code: VAC992V2424** **Level : 100**

Course Objective: The objective of the course **Vernacular Architecture of India (VAC992V2424)** is to introduce students to the fundamental principles of traditional building construction and to equip them with the knowledge necessary to design climatically responsive structures that are rooted in local architectural traditions.

Course Outcomes :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Choose examples of Vernacular Architecture from different regions of India, such as North India, North-East India, South India and West and Central India.	BT1
CO2	Interpret vernacular features across India.	BT2
CO3	Develop insights in traditional construction technique from respective case studies of Vernacular Architecture across India.	BT3
CO4	Analyse differences in Vernacular architecture across India.	BT4

Prerequisite:

- Basic understanding of architectural design principles and construction techniques.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Vernacular Architecture in North-East(Assam and Nagaland)** 1. Vernacular architecture of Assam- Introduction, traditional building material, construction technique and architectural style. 2. Vernacular Architecture of Nagaland- Introduction, Unique feature and cultural significance. 3. Comparison Analyses 4. Conclusion.	5	15
II	**Vernacular Architecture in North India(Himachal Pradesh)** 1. Introduction of Himachal Pradesh, 2. Area location, Introduction to Vernacular Architecture of Himachal Pradesh, 3. Factor influencing vernacular architecture of Himachal Pradesh, 4. Traditional building material and techniques,	5	15

	5. Kath-Khuni Architecture, Settlement Pattern, 6. Examples of Kath-Khuni Houses, Temple and other building, 7. Typologies of Houses, 8. Case studies and 9. Conclusion.		
III	**Vernacular Architecture in West and Central India(Rajasthan)** 1. Introduction to Vernacular Architecture of Gujarat. 2. Area location, Introduction to Vernacular Architecture of Rajasthan. 3. Elements of Vernacular Architecture of Rajasthan 4. Case Study of Rajasthan Houses and other building 5. Conclusion	5	15
IV	**Vernacular Architecture in South India(Kerala)** 1. Introduction to Vernacular Architecture of Kerala. 2. Area location, Introduction to Vernacular Architecture of Kerala. 3. Traditional building material and techniques 4. Typologies of Houses 5. Case studies and 6. Conclusion.	5	15
	TOTAL	**80**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	20 hours	60 hours	10 hours
			Group Work Presentation/ Assignments Field Study

Text Books:

1. Vernacular Architecture of India: Traditional Residential Styles and Spaces (Tejinder S. Randhawa)

Reference Books:

2. Prathaa: Kath Khuni Architecture of Himachal Pradesh

Reference Materials:

1. A study on Living Culture and Typo-Morphology of Vernacular-Traditional Houses in Kerala.
2. Vernacular architecture of Majuli, Assam (2020 IJCRT | Volume 8, Issue 7 July 2020 | ISSN: 2320-2882)

3rd Semester		
Paper 1 Major	**Interior Design I** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Theory + Studio**	**Subject Code: IDS082M311** **Level : 200**

Course Objective: The objective of **Interior Design I (IDS082M311)** is to translate knowledge of design principles through enhancement of thought process into a workable design, apply basic design skills & anthropometric observations by getting involved with progressively difficult design problems.

Course Outcomes :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the characteristics of designing elements and understanding the visual impacts w.r.t furniture's used during civilizations	BT1
CO 2	Outline the basic design understandings, human functions and their implications for space requirements and circulation of space. Also, involve the parameters of design anthropometrics and ergonomics, human activity, particulars of scale, materials etc.	BT2
CO 3	Utilize basic human functions and their implications for space requirements for a proper design.	BT3
CO 4	Analyse an existing space to identify issues, concerns & design potential.	BT4

Prerequisite:

- Basic Concept of Art

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	*P*
I (Theory)	**History of Interior Design:** Study the characteristics of designing elements and understanding the visual impacts w.r.t furniture's used during early Egyptian, Mesopotamian, Greek period, roman period, Middle Ages, early medieval, Romanesque, gothic period. Example: Folding furniture, fitted furniture, wooden furniture, klismos, couches.	15	5
II (Theory)	**Introduction to interior design problem** **Study and Mapping:** Study the parameters of design anthropometrics and ergonomics, human activity & use of interior spaces & furniture. Remember the particulars of scale, anthropometry, material and activity etc. Present them through detailed measured drawings and sketches.	10	30

	Understanding Design of an Individual Space: Basic design understandings, human functions and their implications for space requirements. Minimum and optimum areas for activities/ functions. Understanding user-group & user-profile. Movement and circulation diagrams.		
III (Studio)	**Design of Components of a Residence:** Applying the concepts and study learned to designing components of a residence, Eg: Kitchen, Bathroom, Bedrooms. Applying basic human functions and their implications for space requirements and thus understanding the functions and feasibility for real world application.	10	30
IV (Studio)	**Time Problem – Related to main project: (Timeline – 2-3 days)** Analysis of the existing space. Identify issues, concerns & design potential. Basic human functions and their implications for space requirements. Minimum and optimum areas for mono functions & furniture requirements. Movement and circulation diagrams.	15	5
	TOTAL	120	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture / Tutorial	Studio / Practical	Experiential Learning
4	35 hours	70 hours	15 hours
			(Time from the Lecture Hours to be used for Presentation, Case study and site visit)

Text Books:

1. Time saver standards for building types; De Chiara, Joseph and Crosbie, Michael J.; 2nd Ed.; 2011; Tata McGraw Hill; New Delhi.
2. Neufert Architects Data by Ernst Neufert ; 4th Ed.; 2012; Wiley Blackwell; New Delhi
3. Interior design; Ahmed,A.Kasu; 1st Ed.; 2010; Ashish Book Centre; Mumbai. ISBN8178131862
4. History of Architecture, Sir Banister Fletcher, CBS Publishers & distributors, New Delhi

Reference Books:

1. Ramsay and Sleeper; Architectural graphic standard; 11th Ed.; 2008; John Wiley; New York
2. Paul Laseau, Graphic Thinking for Architects and Designers, John Wiley & Sons.
3. Trewin Copplestone, Arts in Society, Prentice Hall Inc.
4. H. Gardner, Art through Ages.
5. Seetharaman, Premavathy; Interior Design & Decoration; 1st Ed.; 2009; CBS; New Delhi
6. Ching D.K; Architecture: Form, space & order; 1st Ed.; 2007; John Wiley; New York
7. Mary Gilliat Coyran, Interior Design Course, Octopus Ltd., London

Reference Materials:

1. Journal of Interior Design, Volume47, Issue-4.
2. David Fair, Design Graphics, Hodder and Stoughton.
3. Sherril Whiton, Interior Design & Decoration, Prentice Hall

3rd Semester		
Paper 2 Major	**Interior Materials and Construction Techniques-I** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation : Theory + Studio**	**Subject Code: IDS082M312** **Level : 200**

Course Objective: The objective of the course **Interior Materials and Construction Techniques –I (IDS082M312)** is to orient and equip the students with materials and products applied in Interior spaces with relevant construction technology.

Course Outcomes :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic principles of Brick and Stone masonry.	BT1
CO 2	Outline the properties, characteristics, strength, manufacture, processing and application of glass materials.	BT2
CO 3	Utilize basic knowledge of building elements, their function and behavior under various conditions.	BT3
CO 4	Analyze various aspects of Paint and Color scheme in the practical field.	BT4

Prerequisite:

- No prerequisites.

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	*P*
I (Theory)	**Material: Glass** Composition and fabrication of glass, classification, types of glass- wired glass, fiber glass, rock wool, laminated glass, glass concrete blocks - their properties and uses in buildings. Commercial forms available – their physical and behavioral properties, tools and technology of its application in built forms. Material and workmanship, specifications, joinery details of Glass.	15	5
II (Studio)	**Composite masonry** **(Bricks, Stones, Fly Ash, AAC Blocks)** Types of masonry, walls and bonds, buttresses, arches, lintels and coping. Stone types and uses, stone fabrication and installation	10	30
III (Studio)	**Arches** Types of Arches, materials, Design and construction.	10	30
IV (Theory)	**Paints, Coatings and Wall coverings** Characteristics of Paint Materials, coatings, grades, gloss level, Primers, Selection, specification, execution and inspection of the job,	15	5

	Dry Wall, Wall covering materials, installation, protection and Maintenance		
	TOTAL	120	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture / Tutorial	Studio / Practical	Experiential Learning
4	35 hours	70 hours	15 hours
			(Time from the Lecture Hours to be used for Presentation and Market Survey)

Text Books:

1. Building construction; Mackay, W.B. ; 1st Ed.; 2005; Donheed; London.
2. Rangwala, "Building Materials".

Reference Books:

2. Bindra, S.P. andthArora, S.P. Building Construction: Planning Techniques and Methods of Construction, 19 ed. Dhanpat Rai Pub., New Delhi, 2000.
3. Moxley, R. Mitchell"s Elementary Building Construction, Technical Press Ltd.
4. Francis D. Ching, Building Construction Illustrated, Wiley Publishers, 2008.
5. Lisa Godsey, Interior Design Materials and Specifications, Bloomsbury, 2017
6. Chudley, R; Construction technology; 2nd Ed.; 1987; ELBS; Harlow
7. Barry, R; Construction of building; 4th Ed.; 1999; East West Press; New Delhi.

Reference Materials:

1. Choice of Decorative Paints: Recommendation of Interior Designer and Dealers in https://www.ceeol.com/search/article-detail?id=193586
2. The Influence of Building Structure on Human Perception in Interior Spaces in https://eprints.tiu.edu.iq/852/

3rd Semester		
Paper 3 Minor	**History & Theory of Interior Design** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation : Practical**	**Subject Code: IDS082N311** **Level : 200**

Course Objective: The objective of the course **History & Theory of Interior Design (IDS082N311)** is to provide the student of Interior Design knowledge on various developments in Interior design through ages.

Course Outcomes :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the fundamentals of design	BT1
CO 2	Interpret the elements of early Egyptian, Mesopotamian & Greek period.	BT2
CO 3	Apply various design theories.	BT3
CO 4	Analyse the various history and social context of furniture from different eras.	BT4

Prerequisite:

- Basic knowledge of computer.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Practical)	**Fundamentals of Design** Line, plane, volume, symmetry, rhythm, harmony etc.	8	12
II (Theory)	**Positive & Normative Theory** Proxemics theory, privacy theory, assets & liabilities of group design etc. **Early Egyptian, Mesopotamian & Greek period** Characteristic & visual impact of furniture like Folding Furniture, Fitted furniture, Wooden furniture, Characteristic of Mesopotamian Furniture	8	12
III (Theory)	Greek Furniture (Importance – Chairs, Klismos, Couches etc. **Roman Period& Middle Ages**	8	12
IV (Practical)	(Chairs, stools, Couches & others) History and Social context Early Mediaeval, Romanesque, Gothic Furniture	8	12
	TOTAL	**96**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	32 hours	48 hours	24 hours
			Group Work Presentation/ Report Making Office Visits

Text Books:

1. Interior design; Ahmed,A.Kasu; 1st Ed.; 2010; Ashish Book Centre; Mumbai. ISBN8178131862
2. History of Architecture, Sir Banister Fletcher, CBS Publishers & distributors, New Delhi

Reference Books:

1. Seetharaman, Premavathy; Interior Design & Decoration; 1st Ed.; 2009; CBS; New Delhi
2. Ching D.K; Architecture: Form, space & order; 1st Ed.; 2007; John Wiley; New York
3. Mary Gilliat Coyran, Interior Design Course, Octopus Ltd., London

Reading Materials:

1. Sherril Whiton, Interior Design & Decoration, Prentice Hall
2. Joseph De Chiara, McGraw Hill, Time Saver Standards for Interior Design, New York.

3rd Semester		
Paper 4 Interdisciplinary	**Bamboo & Cane crafts of North-East India** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation: Practical**	**Subject Code: IDS082I311** **Level : 200**

Course Objectives: The objective of the course **Bamboo & Cane crafts of North-East India (IDS132G312)** is to expose students to different types of indigenous bamboo and cane crafts of North-east India.

Course Outcomes :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Find the intricate structure and the subtle behaviour of the materials, Bamboo & Cane, as applied to the construction of numerous product categories.	BT1
CO 2	Understand how daily use products, an outcome of a complex interplay of culture & technology are, as understood by the people of North-eastern region of India.	BT2
CO 3	Develop concept of different joineries & different weaving patterns of the material.	BT3
CO 4	Classify and observe local crafts surrounding us.	BT4

Prerequisite:

- Basic concepts of human scale and proportion along with technical drafting.

Detailed Syllabus:

Modules	Topics / Course content	Periods/Hours	
		L	**P**
I	**The Region and the Materials:** Understanding the environment and people of North-east India, growth & distribution of the materials.	**8**	**10**
II	**The Craft: Architectural Elements** Fences, Gates, Bridges & Bullock Carts	**8**	**10**
III	**The Craft: Household Products** Colonial cane furniture & Native Furniture Baskets, Fish traps & Fish baskets, Rain Shields & headgear, fans & some miscellaneous products.	**8**	**10**
IV	**Project Work:** Documentation of Bamboo & Cane products around them/ Portfolio	**8**	**10**
Total		**72**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	32 hours	40 hours	18 hours
			Group Work Presentation/ Report Making Office Visits

Reference/Text Books:

1. *Traditional Wisdom- Bamboo & cane Crafts of North East India:* MP Ranjan, Nilam Iyer, Ghanshyam Pandya.; NID Ahmedabad

3rd Semester		
Paper 5 AEC	**CEN III – Fundamentals of Business Communication** **L-T-P-C : 1-0-0-1 Credits : 1** **Scheme of Evaluation : Theory+Practical**	**Subject Code: CEN982A301** **Level : 200**

Course Objective: The aim of the course is to develop essential business communication skills, including effective writing, speaking, and interpersonal communication, to enhance professional interactions, collaboration, and successful communication strategies within diverse corporate environments.

Course Outcomes: On successful completion of the course the students will be able to:

CO Level	Course Outcome	Blooms Taxonomy Level
CO 1	**Define and list** business documents using appropriate formats and styles, demonstrating proficiency in written communication for various business contexts.	**BT 1**
CO 2	**Demonstrate** confident verbal communication skills through persuasive presentations, active listening, and clear articulation to engage and influence diverse stakeholders.	**BT 2**
CO 3	**Apply** effective interpersonal communication strategies, including conflict resolution and active teamwork, to foster positive relationships and contribute to successful organizational communication dynamics	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	**Business Communication: Spoken and Written** • The Role of Business Communication • Classification and Purpose of Business Communication • The Importance of Communication in Management • Communication Training for Managers • Communication Structures in Organizations • Information to be Communicated at the Workplace • Writing Business Letters, Notice, Agenda and Minutes	**5**

II	**Negotiation Skills in Business Communication** • The Nature and Need for Negotiation o Situations requiring and not requiring negotiations • Factors Affecting Negotiation o Location, Timing, Subjective Factors • Stages in the Negotiation Process o Preparation, Negotiation, Implementation • Negotiation Strategies	**5**
III	**Ethics in Business Communication** • Ethical Communication • Values, Ethics and Communication • Ethical Dilemmas Facing Managers • A Strategic Approach to Business Ethics • Ethical Communication on the Internet • Ethics in Advertising	**5**
IV	**Business Etiquettes and Professionalism** • Introduction to Business Etiquette • Interview Etiquette • Social Etiquette • Workplace Etiquette • Netiquette	**5**

Texts:

1. *Business Communication* by Shalini Verma
2. *Business Communication* by P.D. Chaturvedi and Mukesh Chaturvedi
3. *Technical Communication* by Meenakshi Raman and Sangeeta Sharma

3rd Semester		
Paper 7 SEC	**Computer Application II** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practicum/Jury**	**Subject Code:** IDS082S311 **Level : 100**

Course Objective: The objective of **Computer Application (IDS082S311)** is to develop skills in the Software SketchUp and provide basic and advanced 3D models.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Show computational skills to produce 3D models	BT1
CO 2	Relate drawing aids and equipment along with learning to computerize design drawings	BT2
CO 3	Model an object or space and represent it graphically and digitally	BT3
CO 4	Examine how digital design software is used in the industry for the production of 3D models	BT4

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Interface:** Introduction to the interface, Digital drawings tools, drawing lines and shapes, modifying lines and shapes, drawing with accuracy and speed tools and commands.	4	8
II	**Drawing and Modifying:** Drawing and Modifying objects, properties, Units, dimensions, lines and pen weight. Blocks, group, extrude etc.	4	8
III	**Introduction to Commands:** Basic drawing commands, editing commands, scaling, setting dimensioning variables etc.	4	8
IV	**Presentation and Render:** Analysis of space to represent it as a final 3D model with renders or walkthrough.	4	8
	TOTAL	**48**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	12 hours	32 hours	4 hours
			Group Work Presentation/ Report Making Office Visits

Text Books:

1. SketchUp For Dummies by Bill Fane.
2. SketchUp to LayOut by Matt Donley

Reference Books:

5. SketchUp for Interior Design by L Cline

Reading Material:

1. SketchUp for Builders by John G. Brock.

4th Semester		
Paper 1 Major	**Interior Design II** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio/External**	**Subject Code: IDS082M411** **Level: 200**

Course Objective: The objective of the course **Interior Design II (IDS082M411)** is to familiarize students with Interior Design of a Residence and present the visualized output in 3d format making use of computer applications (Sketchup).

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define adequate facilities for work, relaxation, comfort, privacy, aesthetics, and maintenance through design and proper choice of materials, services, fittings, and fixtures in interiors of residences.	BT1
CO 2	Demonstrate critical thinking while designing a residential space and come up with strategic design solutions.	BT2
CO 3	Utilize the Design Studio to gain hands on experience and understand basic infrastructural requirements	BT3
CO 4	Discover the methods of 3d digital representation in interior design of a residence.	BT4

Prerequisite:

- Basic Knowledge of designing a residence along with drafting and digital presentation techniques.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Studio)	1. Designing of various activities in different rooms of a Residence and factors to be considered while designing. (Living room, Dining, Kitchen/Pantry, Bedroom(s), Balcony /open areas & Toilet(s). 2. Establishment of areas for different units based on function, furniture requirement & number of persons. 3. Orientation, Grouping of user's area, Circulation between and within user's area, Light and Ventilation, Flexibility, Privacy, Roominess (spaciousness), Services, Aesthetics, Requirement for air conditioning & Cost Complete	**6**	26
II (Studio)	1. Floor Layouts, Sectional Elevations, and Furniture Layout of the Different Interior Spaces. Detail layout plan (/tablet plan) of individual space/unit, with 4 sectional elevations and detailed specifications. 2. Design and detailing of furniture and interiors according to consumer ergonomics Application of Colour, Texture &. Pattern through Techniques of Rendering & Presentation. Visual and oral presentation of complete project	6	26

III (Studio)	Introduction to and Familiarization with Sketchup: 1.Interface and basic commands: Introduction to the interface, setting up a drawing template, familiarization with large tool set, scenes, importing .dwg files, saving files, installing extensions, etc. 2. Drawing and Modifying: Drawing and Modifying objects, line, rectangle, circle, arc, object properties, Units, dimensions, lines, and pen weight. Blocks, group, extrude, rotate, add texture, etc. 3.Presentation and Render: Learning analysis of a space to represent it as a final 3D model with renders or walkthrough.	12	20
IV (Studio)	Project: Students to be asked to create a 3D model of the residence in Sketchup and Render the output in software.	12	20
	TOTAL	**128**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	36	92	8 hours
			Group Work Presentation/ Report Making Office Visits

Text Books:

1. De Chiara, Joseph and Crosbie, Michael J. (2011). Time saver standards for building types, 2nd Ed. Tata McGraw Hill, New Delhi.
2. SketchUp For Dummies by Bill Fane.
3. SketchUp to LayOut by Matt Donley.

Reference Books:

1. Neufert, Peter. (2012) Neufert'sArchitects' data, 4th Ed. John Wiley, New Delhi
2. Ramsay and Sleeper. (2008). Architectural graphic standard. 11th Ed. John Wiley, New York.
3. SketchUp for Interior Design by L Cline

Reading Materials:

1. Carlson, C. (2022, December 12). Dezeen's top 10 home interiors of 2022. Retrieved December 14, 2022, from dezeen: https://www.dezeen.com/2022/12/12/top-home-interiors-2022-review/
2. Nabil, S., & Kirk, D. (2021). Decoraction: a Catalogue for Interactive Home Decor of the Nearest-Future. Proceedings of the Fifteenth International Conference Tangible, Embedded, and Embodied Interaction (pp. 1-13). Salzburg: Association for Computing Machinery, New York.
3. Werner, C. M. (1987). Home Interiors: A Time and Place for Interpersonal Relationships. Environment and Behaviour .
4. SketchUp for Builders by John G. Brock.

4th Semester		
Paper 2 Major	**Interior Materials and Construction Techniques-II with Material Specification** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation : Theory + Studio**	**Subject Code: IDS082M412** **Level : 200**

Course Objective: The objective of the course **Interior Materials and Construction Techniques –II with Material Specification (IDS082M412)** is to orient and equip the students with materials and products applied in Interior spaces with relevant construction technology.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic principles of doors, windows, flooring etc.	BT1
CO 2	Outlining the properties, characteristics, strength, manufacture, processing and application of different flooring techniques and materials.	BT2
CO 3	Utilizing basic knowledge of building elements, basically the function and behavior of different types of doors and windows under various conditions.	BT3
CO 4	Analyze various aspects of materials and techniques of each component in the practical field.	BT4

Prerequisite:

- No prerequisites.

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	*P*
I (Theory)	**Timber** ***Material Study:*** Quality of timber used in building, defects, seasoning and preservation of timber, types ; Natural, hard and softwood, uses of timber for aesthetic & structural purposes. ***Material Use:*** Products commonly made of wood, fabrication, finishing processes, protection and maintenance, parts and accessories. Uses of commercial wood in building i.e., plywood, block boards, particleboards, veneers, laminates, MDF, Hard wood, Block Board, WPC board, HDF board and other types.	15	5
II (Studio)	**Doors** ***Construction and its fixtures:*** Types of wooden Doors, i.e., ledged, braced, battened, paneled, flush and glazed doors, study of joinery details.	10	30

III (Studio)	**Windows** ***Construction and its fixtures:*** Types of wooden glazed windows, Fixed, side and top hung pivoted, louvered, ventilators and fanlights, study of joinery details.	10	30
IV (Theory)	**Flooring** ***Introduction:*** General Information. . Flooring techniques, slope difference in flooring, drop slab, tile difference, etc. Type, properties, Sizes, Uses& Limitations. Resilient and Wooden flooring - Use, Properties, Types, Construction, Selection Checklist. ***Flooring materials:*** Different types of flooring materials namely tiles, stone, wood, linoleum, cork, vinyl, rubber, cementitious, Burnt Clay, IPS (Indian Patent Stone Flooring) and terrazzo flooring. Mortars, and Grouts. To study materials uses property. Study of Adhesives: uses and property.	15	5
	TOTAL	120	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture / Tutorial	Studio / Practical	Experiential Learning
4	35 hours	70 hours	15 hours
			(Time from the Lecture Hours to be used for Presentation and Market Survey)

Text Books:

1. Building construction; Mackay, W.B. ; 1st Ed.; 2005; Donheed; London.
2. Rangwala, "Building Materials".

Reference Books:

1. Bindra, S.P. and Arora, S.P. Building Construction: Planning Techniques and Methods of Construction, 19 ed. Dhanpat Rai Pub., New Delhi, 2000.
2. Moxley, R. Mitchell"s Elementary Building Construction, Technical Press Ltd.
3. Francis D. Ching, Building Construction Illustrated, Wiley Publishers, 2008.
4. Lisa Godsey, Interior Design Materials and Specifications, Bloomsbury, 2017
5. Chudley, R; Construction technology; 2nd Ed.; 1987; ELBS; Harlow
6. Barry, R; Construction of building; 4th Ed.; 1999; East West Press; New Delhi.

Reference Materials:

1. Black and Decker, The Complete Guide to Flooring: Design, Planning and Installation for All Types of Flooring, Creative Publishing International, 2003.

The Influence of Building Structure on Human Perception in Interior Spaces in https://eprints.tiu.edu.iq/852/

4th Semester		
Paper 3 Major	**Furnishing, finishing and fittings** **L-T-P-C: 2-0-2-4 Credits: 4** **Scheme of Evaluation: Theory + Practical**	**Subject Code: IDS082M443** **Level: 200**

Course Objective: The objective of the course **Furnishing, finishing and fittings (IDS082M443)** is to design and implement furnishings and finishes and fittings according to the various spaces promoting aesthetical and functional interior design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define basic knowledge of furnishings and finishes and fittings	BT1
CO 2	Demonstrate an ability to understand the importance of furnishings and finishes and fittings in interior design.	BT2
CO 3	Apply theories of furnishings and finishes and fittings of various spaces for design development.	BT3
CO 4	Examine interior furnishings, finishes and fittings and relate them to design an aesthetical and functional interior.	BT4

Prerequisite:

- Basic concepts of color, texture and materials.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Theory)	Wall - different types of wall treatments distemper, emulsion paint, plastic paints textured paint wall paper, tiles cladding, murals and paintings, wains coating (paneling of wall on the lower part of wall to protect from any kind of damage) dado work, wall painting, acoustic wall.	**15**	5
II (Theory)	Flooring - cement polish flour, terrazzo flooring, tiles stone, carpet & rags, mats/jute, woolen, synthetic, straw, vinyl flooring, parqueted /ceiling.	5	15
III (Practical)	Curtains - curtains of different styles, draperies, sheer curtains, blinds, festoons, curtain materials. Upholstery - cotton, synthetic, velvet, brocade, leather synthetic wool, silk, acrylic, polyester – nylon. Cushion- chair cushion, throw pillows, bench cushions, feather cushion, bolster.	5	15

IV (Practical)	Fittings - hinges, handle, magnets, ball catchers, handles, profile, brackets, hydraulic channels, sliding channels, locks, latches, door knocker, door studs. Lights & lightings- need of light, types of lights, the role of proper lighting.	5	25
	TOTAL	**90**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
3	22 hours	60 hours	8 hours
			Presentation/ Report Making Site Visits

Text Books:

1. Designer Lesley Taylor, Series of books of interior design, Series of four books
2. Home lighting ideas, Writer Richard Wiles, Series of four books
3. Jocasta Innes, Published by Ebury Press, National Magazine House, London

Reference Books:

1. The Designer's Guide to Surfaces and Finishes, Radford, Penny

Reference Materials:

1. Interior Design Materials: Important Types Of Materials And Finishes | The Design Gesture
2. Natural Interior Design: How To Add Nature Into Your Homes | The Design Gesture

4th Semester		
Paper 4 Minor	**Interior Design Representation Techniques** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code: IDS082N411** **Level : 200**

Objective: The objective of the course **Interior Design Representation Techniques (IDS082N411)** is to examine the various components of drawing: line, value, texture, composition, and spatial dynamics. Students learn the basics of freehand drawing and get familiarized towards the principles of Design thinking, Visualization and representation with drawing materials and basic equipment.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	How to use basic tools effectively like pencil, pen, etc.	BT1
CO 2	Translate visuals on to drawings.	BT2
CO 3	Utilizing basic knowledge of visualization with understanding of plans, elevation, sections.	BT3
CO 4	Apply visual drawing techniques to create space and depth.	BT3

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Basics of Free hand drawing:** Use of drawing tools and materials. Understanding design element and visual perception Forms analyzed through spatial elements like points, line, curves plane and volume.	5	15
II	**Fonts** Developing writing style of alphabets and numbers for drawings **Scale** Engineering scale and Graphical scale	5	15
III	**Orthographic Projections** Plan, Elevation, Section	5	15
IV	**Perspective Projections** Picture Plane, Vanishing Point, Eye Level, Station Point, Draw simple objects	5	15
	TOTAL	**80**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			True Scale representations, Measure drawing, Exposure to using simple tools.

Text Books:

1. Bhatt, N.D and Panchal, V. M; (2008) Engineering drawing, 1st Ed. Charotar Publishing House; Gujrat.

Reference Books:

1. Scott,R.G (2009); Design Fundamentals; 2nd Ed.; R.E. Kreiger; New York
2. John Montague, Willey, (1985), Basic Perspective Drawing, A Visual Approach, 6th Edition, John Willey and sons, Inc.
3. Francis D.K. Ching, (1998), Design Drawing, John Willey and sons, Inc.
4. Mulik, Shankar, (1994) A Text Book of perspective & Sciography, Allied Publishers Ltd., Mumbai.
5. Sherkey W, Morgan. (1950). Architectural Drawing: Perspective, Light and Shadow, Rendering, Mc Graw Hill.
6. Arthur L. Guptill, Watson. (1997). Rendering in Pen and Ink, Guptill Publications, New York.

Reading Materials:

1. Maria Bako, Different projecting methods in teaching spatial geometry. European Research in Mathematics Education.

4th Semester		
Paper 5 **Minor**	**Material Documentation & Specification** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation : Practical**	**Subject Code:** **IDS082N412** **Level : 200**

Course Objective: The objective of the course **Material Documentation & Specification (IDS082N412)** is to understand the various components of interior spaces and familiarize the students of Interior Design on materials, their specifications and construction methodology.

Course Outcome :

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define knowledge and technicality about different aspects of Flooring, flooring materials used in interior spaces, and implementation/construction of flooring in interior spaces.	BT1
CO 2	Demonstrate technicality about different aspects and types of Ceilings.	BT2
CO 3	Select various External Finishes in the market such as Glazing materials and assembles, coatings and sealants along with waterproofing materials.	BT3
CO 4	Distinguish about different internal Finishes in the market such as coatings, their types and uses and selection checklist.	BT4

Prerequisite:

- Basic knowledge of Interior Materials.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Flooring** Introduction: Type, properties, specifications, Sizes, Uses & Limitations Adhesives - Materials uses property Resilient and Wooden flooring - Use, Properties, Types, Construction	5	15
II	**Ceilings** Introduction: Type, properties, specifications, Sizes, Uses & Limitations Assembly type & Membrane types (Gypsum Boards, Plasters) Suspended ceiling Sound absorbing components	5	15

III	**External Finish- Insulation, Waterproofing/Dam proofing, Exterior wall, Curtain wall, Interior Partition system, Wall facings, Glass/plastics, Windows, Coatings, Sealants** Introduction: Type, Materials, specifications, properties, Sizes, Uses & Limitations Glazing Material- Solar optical properties	5	15
IV	**Internal Finish** Coatings Introduction: Field applied - Type, Materials, specifications, properties, Sizes, Uses& Limitations	5	15
	TOTAL	**80**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			Group Work Presentation/ Report Making Office Visits

Text Books:

1. *"Building Construction"* by W.B.Mackay
2. "*Time Saver Standards for Building Materials &Systems"*Watson Donald

Reference Books:

1. Chudley*"Construction Technology"*
2. Barry*"Construction of Building"* Barry

Reference/Reading Materials:

1. ***Interior Design Magazine***
2. ***ELLE Decor***
3. ***Architonic***
4. *Architectural Digest*
5. *Better Homes and Gardens*
6. *California Home and Design*

4th Semester		
Paper 6 AEC	**CEN IV: Business Communication: Concepts and Skills** **L-T-P-C : 1-0-0-1 Credits : 1** **Scheme of Evaluation : Theory+Practical**	**Subject Code: CEN982A401** **Level : 200**

Course Objectives: This course is designed to enhance employability and maximize the students' potential by introducing them to the principles that determine personal and professional success, thereby helping them acquire the skills needed to apply these principles in their lives and careers.

Course Outcomes: After the successful completion of the course, the students will be able to

CO Level	Course Outcome	Blooms Taxonomy Level
CO 1	**Demonstrate** understanding the importance of verbal and non-verbal skills while delivering an effective presentation.	**BT 2**
CO 2	**Develop** professional documents to meet the objectives of the workplace	**BT 3**
CO 3	**Identify** different life skills and internet competencies required in personal and professional life.	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	**Presentation Skills** Importance of presentation skills, Essential characteristics of a good presentation, Stages of a presentation, Visual aids in presentation, Effective delivery of a presentation	**5**
II	**Business Writing** Report writing: Importance of reports, Types of reports, Format of reports, Structure of formal reports Proposal writing: Importance of proposal, Types of proposal, structure of formal proposals Technical articles: Types and structure	**5**
III	**Preparing for jobs** Employability and Unemployability, Bridging the Industry-Academia Gap Knowing the four- step employment process, writing resumes, Guidelines for a good resume, Writing cover letters Interviews: Types of interview, what does a job interview assess, strategies of success at interviews, participating in group discussions.	**5**
IV	**Digital Literacy and Life Skills** **Digital literacy:** Digital skills for the '21st century', College students and technology, information management using Webspace, Dropbox, directory, and folder renaming conventions. Social Media Technology and Safety, Web 2.0. **Life Skills**: Overview of Life Skills: Meaning and significance of life skills, Life skills identified by WHO: self-awareness, Empathy, Critical thinking, Creative thinking, Decision making, problem- solving, Effective communication, interpersonal relationship, coping with stress, coping with emotion. Application of life skills: opening and operating bank accounts, applying for PAN, Passport, online bill payments, ticket booking, gas booking	**5**

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**

15 hours	-	10 hours - Movie/ Documentary screening - Field visits - Peer teaching - Seminars - Library visits

Text Books:

1. *Business Communication* by Shalini Verma References:
2. *Technical Communication* by Meenakshi Raman and Sangeeta Sharma

5th Semester		
Paper 1 Major	**Interior Design III** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio/External**	**Subject Code: IDS082C511** **Level: 300**

Course Objective: The objective of the course **Interior Design III (IDS132C511)** is to translate knowledge of abstract design principles through enhancement of thought process into a workable design for a 5 star hotel.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the fundamentals of 5 star hotel interior design.	BT1
CO	Summarize design skills and requirements within the constraints of a given interior design project brief.	BT2
CO 3	Make use of the Design Studio to gain hands on experience and understand basic infrastructural requirements	BT3
CO 4	Analyze the space with a design eye and come up with strategic interior design solutions.	BT4

Prerequisite:

- Basic concepts of interior design and drafting.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Studio)	**Field Visit** • Case Study- Five-Star Hotel • Students will conduct a field visit to observe hotel operations and design. • Report on the Visit • Documentation with photographs and observations. • Comparative Analysis	**6**	26
II (Studio)	**Introduction to Hotels** • What are Hotels? • Definition and purpose of hotels. • Categories of Hotels.	6	26

III (Studio)	**Design Development - 1** • Hotel Plan- A floor plan provided for students to work from. • Floor Plan with Furniture Layout • Students will create detailed layouts incorporating furniture arrangements. • Final Drawings (plan, section, elevation, views)	12	20
IV (Studio)	**Design Development – 2 Banquet, Bar and Swimming Pool** • Case Study and material survey • Final Drawings (plan, section, elevation, views)	12	20
	TOTAL	**128**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	36	92	8 hours
			Group Work Presentation/ Report Making Site Visits

Text Books:

1. De Chiara, Joseph and Crosbie, Michael J. (2011). Time saver standards for building types, 2nd Ed. *Tata McGraw Hill*, New Delhi.

Reference Books:

1. Neufert, Peter. (2012) Neufert's architects' data, 4th Ed. *John Wiley, New Delhi*
2. Ramsay and Sleeper. (2008). Architectural graphic standard. 11th Ed. *John Wiley, New York*

Reading Materials:

1. Carlson, C. (2022, December 12). *Dezeen's top 10 home interiors of 2022*. Retrieved December 14, 2022, from dezeen: https://www.dezeen.com/2022/12/12/top-home-interiors-2022-review/
2. Nabil, S., & Kirk, D. (2021). Decoraction: a Catalogue for Interactive Home Decor of the Nearest-Future. *Proceedings of the Fifteenth International Conference Tangible, Embedded, and Embodied Interaction* (pp. 1-13). Salzburg: Association for Computing Machinery, New York.
3. Werner, C. M. (1987). Home Interiors: A Time and Place for Interpersonal Relationships. *Environment and Behaviour* .

5th Semester		
Paper 2 Major	**Interior Materials with Construction Techniques III with Material Specification** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Theory + Studio**	**Subject Code: IDS082M512** **Level: 300**

Course Objective. The objective of the course **Interior Materials and Construction Techniques-III (IDS082M512)** is to orient and equip the students with materials and products applied in Interior spaces with relevant construction technology

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic working of Partitions and materials.	BT1
CO 2	Classify the different types Ceiling Systems and materials.	BT2
CO 3	Utilize the properties, characteristics, strength, manufacture, and application of Finishes.	BT3
CO 4	Analyze various aspects of Panelling in the practical field.	BT4

- **Prerequisite:**
  No prerequisites.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Studio)	**Partition** Wooden Partition with glass fittings Partition with MDF board (CNC Designs) Partition finish with veneers, laminates and other new materials	10	30
II (Theory)	**Ceiling Systems** Introduction to Suspended Ceiling Systems Gypsum Board Ceiling &Modular Ceiling Systems in various materials	15	5
III (Theory)	**Finishes (Interiors and Exteriors)** Terrazos, Composites, Concrete, Laminated Materials- Fabrication, Curing, Finishing (Texture, Pigment, Sealant), Installation, Maintenance	15	5
IV (Studio)	**Panelling** Wall panelling- wood, ply with laminate and veneer, PVC board, MDF board, foam and other new materials related to the present market (Designs with joinery and fixing details)	10	30

	TOTAL	**120**

Notional Credit Hours for the course: 30 x 4= 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	35 hours	70 hours	15 hours
			Model Making, Group Work, Discussions and Presentations, Site Visits.

Text Books:

1. W.B.Mackay;"Building Construction".

Reference Books:

1. Bindra, S.P. andthArora, S.P. Building Construction: Planning Techniques and Methods of Construction, 19 ed. Dhanpat Rai Pub., New Delhi, 2000.
2. Moxley, R. Mitchell‟s Elementary Building Construction, Technical Press Ltd.
3. Francis D. Ching, **Building Construction Illustrated**, Wiley publishers, 2008.
4. Lisa Godsey, Interior Design Materials and Specifications, Bloomsbury, 2017

Reading Materials:

1. Bustillo Revuelta, M. (2021). Terrazzo. In: Construction Materials. Springer Textbooks in Earth Sciences, Geography and Environment. Springer, Cham.

5th Semester		
Paper 3 Major	**Building Services I** **L-T-P-C: 2-0-2-4 Credits: 4** **Scheme of Evaluation: Theory + Practical**	**Subject Code: IDS082M543** **Level: 300**

Course Objective: The objective of **Building Services I (IDS082M543)** is to develop the knowledge and skills required for understanding the water supply and sanitation services in buildings and their integration with architectural design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Relate the technical knowledge regarding water supply and sanitation system in a building.	BT1
CO 2	Demonstrate the principles and best practices for Solid waste management in residential unit.	BT2
CO 3	Apply the principles of sanitation, Introduction to various sanitary pipes, joints, fittings and fixtures, their function, placement and constructional details.	BT3
CO 4	Inference the knowledge into sheet work.	BT4

Prerequisite:

- Basic idea of water sources, supplies, pipes, etc.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Theory)	**Water Supply**: Introduction to water supply, Assessment of water requirements, Sources of supply, collection& method of supply. Catchment areas, reservoirs, recharge of ground water (methods) and their location. Roof top rain water harvesting. Recovery of used water. Physical, Chemical and biological examination of water, Water treatment.	**15**	5
II (Practical)	**Water Management**: Water pipes– materials (GI, PVC, CPVC/ UPVC pipes, introduction to Copper plumbing, Pipe Accessories, Storage tanks, Pumps, Pipes – laying and jointing. Supply for a neighborhood and town. Rainwater Harvesting, Recharging, Recycling and reuse, application in planning, water supply distribution systems (Urban & rural). Sheet work on water supply, overhead water tank and calculations based on present semester design.	5	15
III (Theory)	**Sanitation:** Principles of sanitation, Introduction to various sanitary pipes, joints,	15	5

	fittings and fixtures, their function, placement and constructional details. Study of internal & external drainage system of buildings including small residences, apartments, etc. Single stack system, one pipe and two pipe systems, testing of house drains, Gradients used in laying drains, Self-cleaning and non- scoring velocities for drain pipes Study of Traps, Inspection chambers, Manholes, Septic tanks, Soak pits, Plumbing layout in residential, high raised building.		
IV (Practical)	**Solid waste, management:** Refuse collection, disposal, Incinerator, Composting, Vermi composting, Sanitary Land filling, Trenching etc. Sheet work on septic tank, soak pit, inspection chamber and calculations based on present semester design.	5	15
	TOTAL	**80**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	22 hours	60 hours	8 hours
			Presentation/ Report Making Site Visits

Text Book:

1. *Sanitary Engineering – (Vol I and II);*Deshpande, R. S.; 1 st Ed; Unique Book Cooperation

   Water Supply and Sanitary Engineering; Birdii, G.S; 1st Ed.; 1980; Standard Publishers Distributors

Reference Book:

1. Shah, Charanjit S; *Water Supply and Sanitary Engineering;*
2. Rangwala, S.C; *Water Supply and Sanitary Engineering;* 1st Ed.; 2005; Charotar Publishing House;
3. Fair, G.M, Geyer, J.C. and Okin, D; *Water and Waste water engineering Volume II*; 1st Ed.; 1968; John Wiley & Sons; New York.
4. CPHEEO; *Manual on sewerage and sewerage treatment;* 1st Ed.; 1980; Ministry of works and housing; New Delhi.
5. *Relevant IS Codes of India*

Reference Material:

1. *Renewable energy, basics and technology, supplement volume on integrated energy systems*

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5th Semester		
Paper 4 Minor	**Interior Landscape** **L-T-P-C : 1-0-3-4 Credits : 4** **Scheme of Evaluation : Practicum/Jury**	**Subject Code:** IDS082N511 **Level : 300**

Course Objective: The objective of the course **Interior Landscape (IDS082N511)** is to understand its dynamics and components; and to develop basic skills to deal with interior landscape design projects.

Course Outcome:

After successful completion of the course, student will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Remember the use of abducted and productive reasoning	BT1
CO 2	Understand the principles, theories and models of design thinking	BT2
CO 3	Apply and adopt solution focus strategies.	BT3
CO 4	Analyse and resolve design problems in creative ways.	BT4

Prerequisite:

- No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I (Practical)	**I Introduction to Interior Landscape** • Introduction to Interior Landscape, definitions & related terminologies. *(Natural, Man-made, Rural, Urban)* • Types of Natural Elements *(Stones, Rocks, Pebbles, Water Forms, Indoor plants and Vegetation)* and their visual characteristics *(form, colour, texture, foliage)*. • Man-Made Elements, co-existence of man-made and natural features, case-studies.	5	10
II	**Scales and Typologies of Interior Landscape Proposals (Residential & Commercial Forms).** • Landscape design principles *(based on its natural & made-made elements)* • Layout and Components, Indoor/ Outdoor linkage to spaces. • Landscape design parameters (functional) for Interior spaces, landscape services *(water supply, drainage, lighting etc.)*in Interiors.	5	15
III	**Design with Plants** • Indoor Plants & their functions *(literature study)* • Field identification & documentation of common Indian	5	10

	Trees, Shrubs used in Interiors. Basics of Planting Design &preparation of Planting Manual.		
IV	**Design Studio** • Prepare design brief as per functional requirements, user needs of any Residential or Commercial Interiors. • Proposal and Detail Drawings.	5	25
	TOTAL	**80**	

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
3	22 hours	60 hours	8 hours
			Poster making, Group Work, Debates, Discussions and Presentations

Text Books:

1. Landscape Architecture: Simmonds J.O.; 5 edition (1 March 2013), McGraw-Hill Education.
2. Interior Gardens – Designing and Constructing Green Spaces in Private and Public Buildings: HaikeFalkenberg (2010)
3. Planting Design: Theodore D.Walker, VNR Publications New York.
4. Introduction to Landscape Architecture: Laurie Michael, 2nd Revised edition (1 February 1986), Elsevier Science Ltd.

Reference Books:

1. KrishenPradip; Jungle Trees of Central India; Penguin Random House India Pvt. Ltd, 2013.
2. Grant W. Reid; Landscape Graphics; Whitney Library of Design, Watson Guptill Publications, New York, 2002.
3. Cedric Crocker, All About Landscaping, Ortho Books.
4. Faulkner R & Faulkner S, Inside Today's Home, New York, Holt Rinehart & Winston Inc., 1960
5. Nelson Hammer; Interior Landscape Design; McGraw-Hill,New York, 1991.
6. Charles W Harris and Nicholas T. Dine; Time Saver Standards for Landscape Architecture; 2nd Edition, Mcgraw – Hill, International Edition, Arch. Series, 1988.

6th Semester		
Paper 1 Major	**Interior Design IV** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio/External**	**Subject Code: IDS082M611** **Level: 200**

Course Objective: The objective of the course **Interior Design IV (IDS082M611)** is to familiarize students with Interior Design of a Retail Space and present the visualized output in 3d format making use of computer applications (Auto CAD, Sketchup, Adobe Photoshop).

Course Outcome:

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the fundamentals of retail interior design.	BT1
CO 2	Develop the ability to create customer-focused, brand-aligned retail environments.	BT2
CO 3	Utilize the Design Studio to Be proficient in integrating technology and sustainability into retail design.	BT3
CO 4	Discover methods to present and justify design solutions effectively.	BT4

Prerequisite:

- Basic Knowledge of designing a retail space along with drafting and digital presentation techniques.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Theory)	**Fundamentals of Retail Design** • Introduction to retail spaces and their evolution. • Types of retail formats (boutiques, department stores, kiosks, pop-ups). • Key principles: spatial planning, customer flow, and zoning. • Case study: Analysis of a successful retail space.	**6**	26
II (Theory)	**Branding, Aesthetics, and Visual Merchandising** • Role of branding in retail interiors. • Use of lighting, color, and materials to reflect brand identity. • Basics of visual merchandising: Display strategies and psychology.	6	26

	• Hands-on activity: Create a concept board for a retail brand.		
III (Studio)	**Technology and Sustainability in Retail Design** • Integrating smart technologies (AR, VR, IoT) for an enhanced customer experience. • E-commerce influence on physical store design. • Sustainable design practices in retail spaces: Materials and energy efficiency. • Group activity: Brainstorm innovative sustainable retail concepts.	12	20
IV (Studio)	**Project Development and Presentation** Designing a retail space for a hypothetical brand from the following options: Garment, Leather, Perfume, Jewelry, Kids store, Kitchen and garden equipment, Books, Decorative items, Beauty parlor,Furnishing • Concept, layout, and branding. • Studio-style design sessions with peer feedback. • Final project presentation and critique.	12	20
	TOTAL	**128**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	36	92	8 hours
			Group Work Presentation/ Report Making Site Visits

Text Books:

1. De Chiara, Joseph and Crosbie, Michael J. (2011). Time saver standards for building types, 2nd Ed. Tata McGraw Hill, New Delhi.

Reference Books:

1. Neufert, Peter. (2012) Neufert'sArchitects' data, 4th Ed. John Wiley, New Delhi
2. Ramsay and Sleeper. (2008). Architectural graphic standard. 11th Ed. John Wiley, New York.
3. SketchUp for Interior Design by L Cline

Reading Materials:

1. "Retail Design" by Otto Riewoldt
2. "The New Rules of Retail: Competing in the World's Toughest Marketplace" by Robin Lewis and Michael Dart
3. "Designing Commercial Interiors" by Christine M. Piotrowski and Elizabeth A. Rogers
4. "Detail in Contemporary Retail Design" by Drew Plunkett and Olga Reid
5. "Visual Merchandising and Display" by Martin M. Pegler
6. "Retail Spaces: Small Stores" by Martin M. Pegler
7. "Sustainable Retail Design" by Jerry Yudelson and Ulf Meyer

Supplementary Resources:

- Journals: *Journal of Interior Design*, *Retail Design International*.
- Websites: *Retail Design Institute*, *Dezeen Retail*.
- Case studies: Analysis of renowned retail spaces like Apple Store, Uniqlo, and Muji.

6th Semester		
Paper 2 Major	**Interior Materials and Construction Techniques - IV** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Theory+Studio**	**Subject Code: IDS082M612** **Level: 300**

Course Objective: The objective of the course **Interior Materials and Construction Techniques - IV (**IDS082M612)is to introduce students to the technicalities of building construction and material and to acquaint with various methods of construction.

Course Outcome:

After successful completion of the course, student will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	State basic RCC and related construction illustrations for a proper design.	BT1
CO 2	Summarize the basic design understandings and understand the practicality of design execution.	BT2
CO 3	Apply the different types of structural construct and differentiate the minute details of the same efficiently.	BT3
CO 4	Analyze various aspects of MS and aluminum doors windows, partitions and ceiling in the practical field	BT4

Prerequisite:

- Basic drafting skills.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Studio)	**Introduction to RCC (Basics)** Study the foundation, columns, beams, slab- types and sizes. Also study the basics architectural layout of a building for the RCC part.	6	30
II (Studio)	**Understanding Staircase and its type, Application of various Railing and grill design** Understanding the requirements of staircase. Types of staircases, construction methods of masonry staircase, timber staircase, RCC staircase, steel staircase, study of fire escape staircase. Applying design and joinery details of railing and grill (steel, glass and other materials).	6	30
III (Theory)	**Steel** Analyse properties, architectural and interior design uses of mild steel and stainless steel and carry out a market survey.	22	6

IV (Theory)	**Prevention and Cure** Anti-Termite treatment of Masonry Walls & Woodwork (pre-construction) Water-proofing & Weather-proofing materials (chemical admixture & surface application), Pest Control Treatment, Fire fighting and Thermal Insulation.	22	6
		128	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	56 hours	72 hours	8 hours
			Model making, Group Work, Discussions and Presentations

Text Books:

1. "*Building Construction*". W.B.Mackay
2. *"Engineering materials",* S.C Rangwala – Charotar publishing, Anand 1982

Reference Books:

1. Chudley, "*Construction Technology*"
2. "*Building Construction*". Sushil Kumar
3. Barry *"Construction of Building"* Barry
4. Donald Watson *"Time Saver Standards – Architectural Design Data*"

Reading Materials:

1. Joseph De Chiara "*Time Saver Standards – Interior Design & Space Planning"*
2. E. Allen "*Time Saver Standards – Architectural Detailing*"

6th Semester		
Paper 3 Major	**Building Services II** **L-T-P-C: 2-0-2-4 Credits: 4** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** IDS082M643 **Level: 300**

Course Objective: The objective of **Building Services II (IDS082M643)** is to impart the fundamental concepts, technicalities of building services and to apply the methods in the practicality of the field.

Course Outcome:

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Remember concepts of design, installation, operation and monitoring of the electrical services, lifts and escalators in residential and commercial buildings.	BT1
CO 2	Understand installation techniques of electrical services, lifts and escalators used in building industry.	BT2
CO 3	Apply different techniques and technical knowledge.	BT3
CO 4	Analyze various aspects of operations and monitoring in the practical field.	BT4

Prerequisite:

- Basic idea of electrical fittings, fixtures, lighting types etc.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Theory)	**Electrical system and fittings, (lighting materials and fixtures)** Basics of electricity – Single/Three phase supply. Types and sizes of all electrical fittings: wires, conduits (aluminum metallic, non-metallic), casing, capping, Bus way, Bus bar, lighting track, fans, Tube-light fittings, Lamp fittings, Chandeliers, etc. Power handling, equipment, roses, holders, switches, sockets, switchboards, MCB, ELCB, distribution board, mains, fuse, meter, circuit breaker etc. Earthing for safety – types of earthing. Electrical wiring systems in domestic and commercial buildings. Selection of cable/wire sizes; potential sources of fire hazards and precautions. Installation of Electronic and Communication systems,	15	10
II (Practical)	**Electrical layout in buildings** Basic planning and layout of installations within a building complex. Main and distribution boards – Transformers – switch gears – substations – space requirement. Emergency supply-standby (Generators, invertors) & UPS. A specific design problem on this	10	15

	aspect.		
III (Theory)	**Elevator (Lifts)** Brief history, Types of Elevators and sizes like traction, Hydraulic etc., Double-decker, Sky lobby, lift lobby, lift interiors etc. Definition and components, elevating a building: i.e., location in building, serving floors, grouping, size, shape of passenger car, door arrangement etc. Brief introduction to Service requirements: Quality of service, quantity of service, time, passenger handling capacity, space and physical requirements, machine room spaces.	15	10
IV (Practical)	**Escalators** Escalators- Definition, Application, Location and arrangement in building. Space requirement.	10	15
	TOTAL	**100**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	40 hours	60 hours	20 hours
			Presentation/ Report Making Site Visits

Text Books:

1. Philips, Derek; *Lighting in Architectural Design;* 1st Ed.; 1964; McGraw Hill; New York.
2. Hopkenson, R.G. & Kay, J.D.; *The lighting of Buildings;* 1st Ed.; 1969; Faber & Faber; London.

Reference Books:

1. *National Building Code.*
2. *Electrical systems.*
3. Handbook of building Engineers in metric systems; 1968; New Delhi
4. Ambrose, E.P.; *Electric Heating;* 1st Ed,; 1968; John Wiley& Sons; New York.

Reading Materials:

1. *National Building Code*
2. Elevators, Escalators, Moving Walkways – Manufactures catalogues.

6th Semester		
Paper 4 Major	**Working Drawing - I** **L-T-P-C : 1-0-3-4 Credits : 4** **Scheme of Evaluation : Practicum/Jury**	**Subject Code: IDS082M614** **Level : 300**

Course Objective: The objective of the course **Working Drawing - I (IDS082M614)** is to enable the students to learn the techniques of preparing detailed drawings which are used for execution works for construction of buildings.

Course Outcome:

After successful completion of the course, student will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Show how detailed drawings for construction works are done.	BT1
CO 2	Demonstrate the working principles and theories related	BT2
CO 3	Apply and adopt the working style done on-site.	BT3
CO 4	Analyze and resolve design problems in creative ways.	BT4

Prerequisite:

- Interior Design drawings of Semester III, IV or V.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I (Practical)	**Preliminary drawings for project plans** Drafting and preparing sheets for unit plans / detail plans of earlier design subject- presentation plan, detailed floor plans, sectional elevations, electrical drawings, service drawings, furniture layout.	5	20
II (Practical)	**Detailed drawings for kitchen** Drafting and preparing sheets for kitchen- presentation plan, detailed plans, service drawings, tile layout, other fixture details, sectional elevation.	5	20
III (Practical)	**Detailed drawings for services** Drafting and preparing sheets for toilet- presentation plan, detailed plans, service drawings, tile layout, other fixture details, sectional elevation. Understanding of services about how water supply, sanitation and electricity run in the building and the site.	5	20
IV (Practical)	**Detailed drawings for architectural fixtures** Drafting and preparing sheets for door and window- detailed plans, sections, elevations and fixture details. Drafting and preparing sheets for furniture design- detailed plans, sections, elevations and fixture details.	5	20
	TOTAL	**100**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	20 hours	80 hours	20 hours
			Site visit, Market survey, Conduct rate analysis.

Text Books:

8. Ching, F. D. (2020). *Building construction illustrated*. John Wiley & Sons.
9. Bizley, G. (2010). *Architecture in detail II*. Routledge.

Reference Books:

3. Allen, E., & Rand, P. (2016). *Architectural detailing: function, constructibility, aesthetics*. John Wiley & Sons.
4. Hall, D. J., & Giglio, N. M. (2016). *Architectural Graphic Standards*. John Wiley & Sons.

Reading Materials:

6. De Chiara, J., Panero, J., & Zelnik, M. (2001). *Time-saver standards for interior design and space planning*. McGraw-Hill.
7. Calloway, S., & Powers, A. (1991). The elements of style: an encyclopedia of domestic architectural detail. *(No Title)*.
8. McLeod, V. (2010). Detail in contemporary timber architecture. *(No Title)*.
9. Emmitt, S. (2023). *Barry's advanced construction of buildings*. John Wiley & Sons.
10. Silver, P., Silver, P., & McLean, W. (2013). *Introduction to Architectural Technology Second Edition*. Hachette UK.

6th Semester		
Paper 5 Minor	**Vastu In Interior Design** **L-T-P-C : 1-0-3-4 Credits : 4** **Scheme of Evaluation : Practicum/Jury**	**Subject Code:** IDS082N641 **Level : 100**

Course Objective: The objective of the course **Vastu in Interior Design (IDS082N641)** is to develop cognitive, strategic and practical thinking and ideation processes by which vastu concepts are developed.

Course Outcome:

After successful completion of the course, student will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Remember the use of abducted and productive reasoning	BT1
CO 2	Understand the principles, theories and models of design thinking	BT2
CO 3	Apply and adopt solution focus strategies.	BT3
CO 4	Analyze and resolve design problems in creative ways.	BT4

Prerequisite:

- No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Vaastu Shastra** Overview of Vaastu Shastra, Vaastu Purusha Mandala, Cosmic Directions, Design Concepts, Experiences of Space and Form, Cardinal and Ordinal Directions, Vaastu and its relation to Human Beings.	**10**	4
II	**Philosophies, Practices, and Principles of Vaastu** Understanding various Philosophies, Practices and Principles of Vaastu and its importance in Practical Aspects , Discussions and Analysis of Vaastu Principles, Positive and Negative Effects, etc.	10	15
III	**Application of Vaastu Principles** Vaastu for Residential Buildings, Apartments, Commercial Buildings, etc., Understanding importance of colour, lighting, material aspects and its effects, Deviation of Degrees, Direction of Roads, Location of Bedrooms, Kitchen, Puja Room, Living Room, Toilets etc., Closed corners, Raised Spaces etc.	10	25
IV (Studio)	**Layout Planning** Detailed Interior Layout Plan of Residential & Commercial Buildings by applying Vaastu Principles.	6	30
	TOTAL	**110**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	36 hours	74 hours	10 hours
			Poster making, Group Work, Debates, Discussions and Presentations

Text Books:

1. Harry N. Abrams (1 October 2007), ISBN-13: 978-1584796398,. Space Matters: Use the Wisdom of Vastu to Create a Healthy Home. 11 Top Designers Show You How.
2. Sokolova, O. (2017). Your Happy House: Illustrated Vastu Shastra for Everyone.
3. Ananth, S. (1998). The Penguin Guide Book to Vaastu. Penguin Group.
4. Bansal, A. K. (2002). Vastu: How to Create a Harmonious Home Through Ancient Indian Design.

Reference Books:

1. Dass, S. S. (2013). The Miracles of Vaastu Shastra .
2. Pandit, S. (2004). Golden Rules Of Vastu Shastra - Remedies And Solutions. Ubs Publishers' Distributors (P) Ltd.
3. Manjul Publishing House Pvt. Ltd. (1 March 2002),, Vaastu Niwas, ISBN-13: 978-8186775127

7th Semester		
Paper 1 Major	**Interior Design V** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio/External**	**Subject Code: IDS082M711** **Level: 400**

Course Objective: The objective of the course **Interior Design V (IDS082M711)** is to familiarize students with basics of designing for health care spaces and environments and to develop skills required for the same on both large- and small-scale spaces with an emphasis on planning and designing compatible spaces for health care services.

Course Outcome :

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the fundamentals of health care interior design.	BT1
CO 2	Develop the ability to create a space with a design eye and come up with strategic design solutions	BT2
CO 3	Utilize the Design Studio to Be proficient in integrating technology and sustainability into health care design with adequate facilities for work, relaxation, comfort, privacy, aesthetics, and maintenance through design and proper choice of materials, services, fittings and fixtures in a designed interior space.	BT3
CO 4	Discover methods to present and justify design solutions effectively.	BT4

Prerequisite:

- Basic Knowledge of spaces of health care buildings along with drafting and digital presentation techniques.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction of Design Fundamentals & Measured drawing** • Introduction of the design project • Comprehension of spatial units • Use of design principles, anthropometry & ergonomics • Fundamentals of Universal Design, acoustics, lighting schemes, colour concepts & use of materials. Measured drawing layout of the space for the proposed interior.	**6**	26
II	**Design Development of Spaces (Foyer, lobby, waiting areas, OPD, Rooms, Services etc.)** • Theme	6	26

	• Schematic & Conceptual Drawings, • Colour scheme • Signages • Final Drawings (plan, section, elevation, views)		
III	**Design Development w.r.t lighting and furniture arrangements of ICU's and OT's.** • Theme • Schematic & Conceptual Drawings • Colour scheme • Specification of materials used • Specific requirements. • Final Drawings (plan, section, elevation, views)	12	20
IV	**Time problem** **Final detailed drawings and Portfolio submission** • As per the requirement of the concerned Teacher	12	20
	TOTAL	**128**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	36	92	8 hours
			Group Work Presentation/ Report Making Site Visits

Text Book:

1. *Interior Design: The New Freedom*, Diamonstein, Barbaralec*;* 1st Ed.; 1982; Rizzoli International Publications; New York.

Reference Books:

1. Diamonstein, Barbaralec; *Interior Design: The New Freedom;* 1st Ed.; 1982; Rizzoli International Publications; New York.
2. Poore, Jonathan; *Interior Colour by Design;* 1st Ed.; 1994; Rockport Publishers;
3. Sha, Rikuyo; *Worldwide Interiors: International Federation of Interior Architects & Designers;* 1987; Japan.

Reading Materials:

7th Semester		
Paper 2 Major	**Research Methodology** **L-T-P-C: 4-0-0-4 Credits: 4** **Scheme of Evaluation: Theory**	**Subject Code: IDS082M703** **Level: 400**

Course Objective: The objective of the course **Research Methodology (**IDS082M703**)** is to introduce the students to the field of academic research and enable them to establish a strong research foundation grounded in both theory and practice. And, to infuse logical reasoning to achieve clarity of thought and also to orient the students to structured research in a focused manner. The process of study shall enable students to conduct in depth analysis and objective research on a topic of their interest. Students may be encouraged to select the topic which may eventually culminate in the Interior Design Thesis.

Course Outcome:

After successful completion of the course, students will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define key research concepts, including types of research, approaches, and distinctions between design studio, dissertation, and thesis with respect to interior design.	BT1
CO 2	Demonstrate the knowledge of different components of research proposal and appropriate research tools and techniques for data collection and analysis.	BT2
CO 3	Organize literature review and case studies effectively to identify research gaps.	BT3
CO 4	Analyze and apply different data interpretation techniques to effectively report research findings using structured formats and prepare comprehensive design-based research documents.	BT4

Detailed Syllabus:

Modules	Course content	Periods/Hours	
		L	P
I	Introduction to Research in Architecture: • Meaning of research, Types of research, Different Approaches to Research (Quantitative, Qualitative, Quasi) • Difference between design studio, dissertation and design thesis. • Formulation of Research Proposal: Aims, Objectives, Methodology, Scope and Limitations • Difference between research question and hypothesis	28	0

II	Tools and Methods: • Selection of Research Topic, Research Database Search, Bibliometric Analysis, Literature Review and Case studies • Scientific methods of research with special emphasis on design research methods • Design enquiry, Visual observations, Primary and secondary data • Data Collection and Survey techniques, Sampling, Questionnaire types • Data analysis techniques, Visual interpretation of data	28	0
III	Presentation Techniques: • Formats for presentation of data, case studies and analysis. • Formats for presentation of thesis design- media appropriate in the interior design profession such as two-dimensional drawing, physical models, and three-dimensional computer models.	28	0
IV	Report Writing/ Paper Writing: • Techniques in report/paper writing • Checklist for Journal Paper Writing vs. Thesis Report Writing • Synopsis and abstract formulation. • Presentations of relevant contextual information, interpretation of the data collected and design; reporting the design development from concept to design solution, explaining the relation of the design to existing knowledge on the topic in the form of coherently written matter. • Introduction to plagiarism: Citation, Referencing, Bibliography using Online Referencing tools, annexure etc.	28	0
TOTAL		**112**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	112	0	8 hours
			Group Work Presentation/ Report Making Site Visits

Text Books:

1. Booth, Wayne C., Colomb, Gregory G. and Williams, Joseph M. (2003), "*The Craft of Research*", Second Edition. Publishers: The University of Chicago Press, Chicago.

2. Vaux, D.E., & Wang, D. (Eds.). (2020). Research Methods for Interior Design: Applying Interiority (1st ed.). Routledge. https://doi.org/10.4324/9780429029325

Reference Books:

1. Sommer, Robert & Sommer, Barbara (2002), "A Practical Guide to Behavioural Research: Tools and Techniques". New York: The Oxford University Press.
2. Creswell, John W. (2003), "*Research Design: Qualitative, Quantitative, and Mixed Method Approaches*". 2nd Edition. Thousand Oaks, CA: Sage Publications.
3. Borden, I. and Ray, K. R. (2006), "*The Dissertation: An Architecture Student's Handbook.* Second Edition. Oxford: Architectural Press.
4. Anderson, J. and Poole, M. (1998), "*Thesis and Assignment Writing*". Brisbane: John Wiley

7th Semester		
Paper 3 Major	**Project Management and Estimation, Costing** **L-T-P-C: 4-0-0-4 Credits: 4** **Scheme of Evaluation: Theory**	**Subject Code:** IDS082M703 **Level: 400**

Course Objective: The objective of the course **Project Management and Estimation, Costing (IDS082M703)** is to equip students with essential skills in project planning, execution, and management, along with accurate estimation, costing, and tendering. Students will learn industry-standard scheduling, budgeting, resource allocation, and risk assessment techniques. They will also develop the ability to prepare detailed cost estimates, specifications, and tender documents for construction and interior projects, ensuring efficient execution and financial planning. to enable the students to learn the techniques of preparing detailed drawings which are used for execution works for the construction of buildings.

Course Outcome:

After successful completion of the course, students will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the project management fundamentals, including project management stages to enhance professional competency in interior design.	BT1
CO 2	Demonstrate the knowledge of Project Monitoring which will enhance in assessing the costs and risks of projects.	BT2
CO 3	Organize and define the estimation terminologies and explain the methods for accurate estimation.	BT3
CO 4	Analyze and apply the estimation methods to determine project costs and study the tendering and bidding process	BT4

Prerequisite:

- Basic understanding and knowledge of Design and construction process of a Project
- Basic knowledge of mathematical calculations.

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I (Theory)	**Fundamentals of Project Management, Project Management Stages** Need for Project planning, scheduling & management. Method of planning & programming. Role of Stakeholders in projects Work breakdown structure, Life cycle of a project. Qualities of an ideal organization Project initiation, planning and scheduling phases	28	0

	Use of management techniques- Bar Chart, Mile stone Chart CPM network analysis &PERT time analysis, time computation & network analysis.		
II (Theory)	**Project Monitoring, Time reduction and Optimization** Project cost, Indirect project cost, Direct project cost, Slope of the direct cost curve. Project Cost Analysis; Risk Analysis Tools for monitoring and control Project closing; Pre-termination of project Project completion audit	28	0
III (Theory)	**Introduction to Estimation, Approximate and Detailed Estimation** Importance of Estimation, Costing and Tendering. Types of measurements, modes of measurements: methods of taking out quantities, preparation of schedule or bill of quantities. Load bearing and RCC framed structure cost using measurement book, abstract sheet, PWD rates to find detailed estimation.	28	0
IV (Theory)	**Rate Analysis, Contract Management and Tendering** Rate analysis of various items of work: preparation of various items of work in the interior works. Detailed rate analysis of Interior items as per current market rates for information only. Percentage of various materials used in building interior items like Plywood, Putty, Tiles, Metal, Laminates, Paints, etc. Technical bid and price bid.Contract types and documents Pre-Contract stage; Post Contract Stage,Types of Tender documents, e-Tendering, FIDIC	28	0
	TOTAL	**112**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	112 hours	0 hours	8 hours
			Site visit, Market survey, Conduct rate analysis.

Text Books:

1. Gopalakrishnan P., Rama Moorthy V.E., (2014), "*Textbook of Project Management*", Trinity Press
2. "Estimating and Costing" by S K Dutta.
3. "Estimating" by SC Rangwala.

Reference Books:

1. Punmia B.C., Khandelwal K.K., (2013), "*Project Planning and Control with PERT and CPM*", Laxmi Publications (P) Ltd.
2. Jerome D.Wiest and Ferdinand K.Levy, *A Management Guide to PERT, CPM*, prentice Hall of India Pub, Ltd., New Delhi, 1982
3. 2. R.A. Burgess and G.White, *Building production and project Management*, The construction press, London, 1975.

Reading Materials:

1. Scrum: The Art of Doing Twice the Work in Half the Time by Jeff Sutherland and J.J. Sutherland
2. A Guide to the Project Management Body of Knowledge by Project Management Institute
3. Project Management for The Unofficial Project Manager by Kory Kogon, Suzette Blakemore, & James Wood
4. Fundamentals of Project Management by Joseph Heagney
5. Making Things Happen: Mastering Project Management by Scott Berkun
6. Rescue the Problem Project: A Complete Guide to Identifying, Preventing, and Recovering from Project Failure by Todd Williams.
7. PWD / CPWD rates as per applicable year.

7th Semester		
Paper 4 **Major**	**Working Drawing - II** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation : Practical**	**Subject Code:** **IDS082M714** **Level : 400**

Course Objective: The objective of **Working Drawing - II (**IDS082M714**)** is to equip students with the skills to produce construction-ready drawings that facilitate the building process.

Course Outcome:

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the technical details of a project through precise and accurate drawings that facilitate effective implementation by workers on site.	BT1
CO 2	Demonstrate the technical base required for the execution of the designer's ideas, ensuring a seamless translation from concept to reality.	BT2
CO 3	Select and create technical drawings that effectively communicate design intent to workmen, enabling accurate execution and realization of the design.	BT3
CO 4	Distinguish the technical knowledge through precise and informative presentation drawings that effectively communicate design intent and details..	BT4

Prerequisite:

- Interior Design drawings of Semesters III, IV, V & VI. Well-versed in Working Drawing - I from 6th Semester.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Detail Drawing of Residential/ Apartment** To produce detailed drawings of a previously designed residence/apartment, focusing on the design and layout of interior elements like bed and bedhead, wardrobe, cabinet shelves and showcase. Dimensions and specifications of each element along with material specifications and finishes.	4	24
II	**Detail Drawing of Commercial Spaces** To produce detailed drawings of various commercial spaces,	4	24

	focusing on the design and layout of interior elements, furniture, and equipment. Office, bookstores, commercial buildings, Workstations, Executive cabins, Conference rooms. Dimensions and specifications of each space, Design and layout of furniture, fixtures, and equipment (FF&E).		
III	**Detail Drawing of Hotel Spaces** To produce detailed drawings of Reception Area, Reception Table, and Lobby Furniture Design, focusing on dimensions, materials, and design considerations to create functional and aesthetically pleasing hotel spaces.	4	24
IV	**Detail Drawing of Amenity Spaces** To produce detailed drawings of various amenity spaces, focusing on the design and layout of interior elements, furniture, and equipment. Conference Room Board Room Common Room Dimensions and specifications of each space Design and layout of furniture, fixtures, and equipment (FF&E). Incorporation of audio-visual and technological requirements.	4	24
	TOTAL	**112**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	16 hours	96 hours	8 hours
			Site visit, Market survey, Conduct rate analysis.

Text Books:

1. Ching, F. D. (2020). *Building construction illustrated*. John Wiley & Sons.
2. Bizley, G. (2010). *Architecture in detail II*. Routledge.

Reference Books:

1. Allen, E., & Rand, P. (2016). *Architectural detailing: function, constructability, aesthetics*. John Wiley & Sons.
2. Hall, D. J., & Giglio, N. M. (2016). *Architectural Graphic Standards*. John Wiley & Sons.

Reading Materials:

1. De Chiara, J., Panero, J., & Zelnik, M. (2001). *Time-saver standards for interior design and space planning*. McGraw-Hill.
2. Calloway, S., & Powers, A. (1991). The elements of style: an encyclopaedia of domestic architectural detail. *(No Title)*.

3. McLeod, V. (2010). Detail in contemporary timber architecture. *(No Title)*.
4. Emmitt, S. (2023). *Barry's advanced construction of buildings*. John Wiley & Sons.
5. Silver, P., Silver, P., & McLean, W. (2013). *Introduction to Architectural Technology Second Edition*. Hachette UK.

7th Semester		
Paper 5 Minor	**Furnishing, finishing and fittings** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Practicum/Jury**	**Subject Code:** IDS082N715 **Level: 400**

Course Objective: The objective of the course **Furnishing, finishing and fittings (IDS082N715)** is to design and implement furnishings and finishes and fittings according to the various spaces promoting aesthetical and functional interior design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define basic knowledge of furnishings and finishes and fittings	BT1
CO 2	Demonstrate an ability to understand the importance of furnishings and finishes and fittings in interior design.	BT2
CO 3	Apply theories of furnishings and finishes and fittings of various spaces for design development.	BT3
CO 4	Examine interior furnishings, finishes and fittings and relate them to design an aesthetical and functional interior.	BT4

Prerequisite:

- Basic concepts of color, texture and materials.

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I (Practical)	Wall - different types of wall treatments distemper, emulsion paint, plastic paints textured paint wall paper, tiles cladding, murals and paintings, wains coating (paneling of wall on the lower part of wall to protect from any kind of damage) dado work, wall painting, acoustic wall.	**20**	10
II (Practical)	Flooring - cement polish flour, terrazzo flooring, tiles stone, carpet & rags, mats/jute, woolen, synthetic, straw, vinyl flooring, parqueted /ceiling.	5	15
III (Practical)	Curtains - curtains of different styles, draperies, sheer curtains, blinds, festoons, curtain materials. Upholstery - cotton, synthetic, velvet, brocade, leather synthetic wool, silk, acrylic, polyester – nylon. Cushion- chair cushion, throw pillows, bench cushions, feather cushion, bolster.	5	15

IV (Practical)	Fittings - hinges, handle, magnets, ball catchers, handles, profile, brackets, hydraulic channels, sliding channels, locks, latches, door knocker, door studs. Lights & lightings- need of light, types of lights, the role of proper lighting.	10	30
	TOTAL	**110**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	40 hours	70 hours	10 hours
			Presentation/ Report Making Site Visits

Text Books:

4. Designer Lesley Taylor, Series of books of interior design, Series of four books
5. Home lighting ideas, Writer Richard Wiles, Series of four books
6. Jocasta Innes, Published by Ebury Press, National Magazine House, London

Reference Books:

2. The Designer's Guide to Surfaces and Finishes, Radford, Penny

Reference Materials:

3. Interior Design Materials: Important Types Of Materials And Finishes | The Design Gesture
4. Natural Interior Design: How To Add Nature Into Your Homes | The Design Gesture

8th Semester		
Paper 1 Major	**Detailed Project Report L-T-P-C: 2-0-2-4 Credits: 4 Scheme of Evaluation : Practical**	**Subject Code: IDS082M811 Level : 400**

Course Objective: The objective of the subject **Detailed Project Report (IDS082M811)** is to enable students to systematically prepare a comprehensive Detailed Project Report (DPR) for their thesis project by integrating design, technical documentation, cost estimation, project planning, sustainability considerations, and implementation strategies required for professional practice and project execution.

Course Outcome:

After successful completion of the course, student will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the components, structure and documentation requirements of a Detailed Project Report.	BT1
CO 2	Explain project planning, design development, technical documentation and feasibility considerations required in a DPR.	BT2
CO 3	Prepare project documentation, estimates, schedules and technical reports for a thesis project.	BT3
CO 4	Analyze the technical, financial, environmental and implementation aspects of a project to develop a comprehensive DPR suitable for professional practice.	BT4

Prerequisite:

- Basic idea about Estimation costing and Project Management

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Project Identification, Literature Study and DPR Framework** • Introduction to Detailed Project Reports (DPR): Purpose, scope and components. • Identification of project vision, goals and objectives. • Literature review and precedent studies related to the thesis project. • Site profile, location analysis and project background. • User needs assessment and stakeholder analysis. • Project brief formulation and space/program requirements. • Preparation of DPR structure, chapter organization and documentation methodology.	10	15

II	**Project Planning and Design Documentation** • Development of conceptual framework based on thesis design. • Design philosophy, planning principles and project rationale. • Preparation of site plan, zoning diagrams and circulation studies. • Architectural/interior planning documentation. • Design development drawings, schematic layouts and visualizations. • Regulatory requirements, codes, standards and accessibility guidelines. • Sustainability, climate responsiveness and environmental considerations.	10	15
III	**Technical Documentation, Costing and Project Management** • Preparation of detailed drawings and specifications. • Material selection and justification. • Structural, electrical, plumbing, HVAC and service integration overview. • Quantity estimation and Bill of Quantities (BOQ). • Cost analysis and project budgeting. • Project scheduling using Gantt Charts/PERT/CPM techniques. • Risk assessment, maintenance considerations and quality management strategies.	10	15
IV	**DPR Compilation, Evaluation and Final Submission** • Compilation of complete DPR document. • Executive summary and project justification. • Feasibility analysis: technical, economic, social and environmental. • Implementation strategy and phasing plan. • Preparation of graphical presentations, infographics and report layouts. • Review, editing and validation of report contents. • Final DPR submission, presentation and viva-voce.	10	15
	TOTAL	**100**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	40 hours	60 hours	20 hours
			Field Visits, Market Survey, Documentation Screening

Text Book:

1) Akshay S. Kamat & Pravin V. Kulkarni – *Construction Project Management*; McGraw Hill Education.
2) Gopalakrishnan, P. – *Project Management*; Macmillan Publishers India Ltd.
3) Prasanna Chandra – *Projects: Planning, Analysis, Selection, Financing, Implementation and Review*; McGraw Hill Education.
4) B.C. Punmia, K.K. Khandelwal & Ashok Kumar Jain – *Estimating, Costing and Valuation*; Laxmi Publications.
5) Joseph De Chiara, Julius Panero & Martin Zelnik – *Time-Saver Standards for Interior Design and Space Planning*; McGraw-Hill.

Reference Books:

1) Architect's Handbook of Professional Practice – American Institute of Architects (AIA).
2) Project Management: A Systems Approach to Planning, Scheduling, and Controlling – Harold Kerzner.
3) Project Planning, Scheduling and Control – James P. Lewis.
4) Building Construction Illustrated – Francis D.K. Ching.
5) Interior Design Illustrated – Francis D.K. Ching.
6) Project Management Body of Knowledge (PMBOK® Guide) – Project Management Institute (PMI).
7) Construction Cost Estimating for Project Control – For advanced costing and project control techniques.
8) **National Building Code (NBC) of India** – Latest Edition, Bureau of Indian Standards (BIS).
9) **CPWD Works Manual** – Latest Edition, Central Public Works Department.
10) **Development Control Regulations and Local Building Bye-laws** applicable to the project location.

8th Semester		
Paper 3 Major	**Thesis / Major Project** **L-T-P-C: 0-0-12-12 Credits: 12** **Scheme of Evaluation: Studio/External**	**Subject Code: IDS082M813** **Level: 400**

Course Objective: The objective of the course **Thesis / Major Project (IDS082M813**) is to translate knowledge of abstract design principles through enhancement of thought process into a workable design and showcasing that for a final complete project.

Course Outcome:

After successful completion of the course, student will be able to -		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define adequate facilities for work, relaxation, comfort, privacy, aesthetics, and maintenance through design and proper choice of materials, services, fittings and fixtures in interiors.	BT1
CO 2	Develop design skills within the constraints of a given interior design project and showcase its complete design language to completion	BT2
CO 3	Utilize the time to showcase hands on experience and understanding of basic infrastructural requirements	BT3
CO 4	Analyze critically with a design eye and come up with strategic design solutions	BT4

Prerequisite:

- Proper concepts of design and drafting.
- Proper knowledge and understanding of Interior Design.

Detailed Syllabus:

Modules	Topics & Course Contents	Periods
I	**Synopsis, Case Study, Site Analysis and Area Programming:** • Synopsis - Brief introduction of the proposed thesis / project. • Literature study and Case Studies - Live & literature, to form a basis for their own design. • Site Analysis - Record and evaluate information on the site and its surroundings, and to use this evaluation in the design response. Area Analysis and Programme - Comparative statement of the various available design standards, areas provided in the various case studies and the area requirements, so that the area requirements for the various functions / spaces for the proposed project can be finalized.	72
II	**Schematic Design:**	72

	• Generate ideas on thebasis of the studies (case studies / literature studies / area analysis) conducted so far in the form of conceptual drawings, sketches and models. Basic concept explaining the principal ideas / thought process for the project in terms of planning / built form / massing of different components, leading to the design, through sketches / 3D images / block models etc	
III	**Design Detailing:** • The schematic drawings presented in the previous module needs to be detailed out as per the comments/ suggestions received from the guides and the reviewers. The detailed drawings as per the final area programme with due consideration to structural and service requirements of the project needs to be prepared.	72
IV	**Design Finalization:** • Pre-Final Design - Final drawings, views, models, etc. incorporating the comments received in the previous reviews. Final Thesis Submission – Submission of all the submittals (drawings, views, models, report, etc.) complete in all respects as per the comments and suggestions received from thesis guide and various review members.	72
	TOTAL	288

Notional Credit Hours for the course: 30 x 12 = 360

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
12	0	288	72 hours
			Field Visit/Sheet Presentation/ Report Making Site Visits/ Seminars

Text Books:

1. De Chiara, Joseph and Crosbie, Michael J. (2011). Time saver standards for building types, 2nd Ed. *Tata McGraw Hill*, New Delhi.
2. As per requirement or suggested by guide.

Reference Books:

1. Neufert, Peter. (2012) Neufert's architects' data, 4th Ed. *John Wiley, New Delhi*

2. Ramsay and Sleeper. (2008). Architectural graphic standard. 11th Ed. *John Wiley, New York*
3. As per requirement or suggested by guide.

Reference Materials:

1. Carlson, C. (2022, December 12). Dezeen's top 10 home interiors of 2022. Retrieved December 14, 2022, from dezeen: https://www.dezeen.com/2022/12/12/top-home-interiors-2022-review/
2. Nabil, S., & Kirk, D. (2021). Decoraction: a Catalogue for Interactive Home Decor of the Nearest-Future. *Proceedings of the Fifteenth International Conference Tangible, Embedded, and Embodied Interaction* (pp. 1-13). Salzburg: Association for Computing Machinery, New York.
3. Werner, C. M. (1987). Home Interiors: A Time and Place for Interpersonal Relationships. *Environment and Behaviour* .

8th Semester		
Paper 3 Minor	**Minor Project** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Practicum/Jury**	**Subject Code: IDS082N812** **Level: 400**

Course Objective: The objective of the course **Minor Project (IDS082N812)** is to develop proficiency in computer-aided architectural and interior design by imparting knowledge of AutoCAD and SketchUp software. It enables students to create, modify, annotate, and present technical drawings and 3D models while applying principles of space planning, ergonomics, aesthetics, and residential interior design. The course also focuses on developing visualization, drafting, and presentation skills required for professional design practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and identify the basic concepts, interface components, drawing settings, commands, layers, and drafting standards used in AutoCAD and SketchUp.	BT1
CO 2	Explain the functions and applications of drawing, modification, annotation, dimensioning, plotting, and visualization tools used in computer-aided architectural and interior design.	BT2
CO 3	Apply AutoCAD and SketchUp tools to create, modify, annotate, and present architectural drawings including plans, sections, elevations, furniture layouts, and interior design compositions.	BT3
CO 4	Analyze residential space requirements, circulation, ergonomics, aesthetics, lighting, ventilation, and functional relationships to develop effective interior layouts and design solutions.	BT4

Detailed Syllabus:

Modules	Course content	Periods/Hours	
		L	P
I	**Introduction to basics of AutoCAD:** Familiarization with AutoCAD interface. Setting up drawing: standard units. Introduction to drawing properties tab. Setting up layers. Create and Save AutoCAD drawing files Define Positions using the Basic Entry methods. Learning to creating basic drawings in AutoCAD. Introduction to use of various commands used for drawing creation:	7	21

	LINE. PLINE, REC, CIRCLE, ARC, OSNAP, etc. Introduction to use of various commands used for drawing modification: MOVE, COPY, TRIM, DIVIDE, OFFSET, CHAMFER, EXPLODE, ERASE, etc. Creation of a composition using basic geometrical shapes in AutoCAD.		
II	**Drawing of plan, section and elevation of a small residence in AutoCAD.** i. Drawing of walls using LINE, PLINE, SPLINE, etc. ii. Creation of door and window using TRIM, CIRCLE, etc. iii. Creation of floor tiles using HATCH. iv. Creation of labels using MTEXT. v. Adding dimension using DIMENSION, DIMSTYLE. vi. Drawing and adding furniture using BLOCK, IMPORT. Learning to print (plot) a drawing in AutoCAD: Page layout for printing and Viewport. Introduction to page setup manager. Setting required paper size and printer (plotter) for printing (A1, A2, A3). Setting Viewport margins and assigning plot area. Setting Plot scale, plot style table and paper orientation.	7	21
III	**Designing of various activities in different rooms of a Residence and factors to be considered while designing.** (Living room, Dining, Kitchen/Pantry, Bedroom(s), Balcony /open areas & Toilet(s). Establishment of areas for different units based on function, furniture requirement & number of persons. Orientation, Grouping of user's area, Circulation between and within user's area, Light and Ventilation, Flexibility, Privacy, Roominess (spaciousness), Services, Aesthetics, Requirement for air conditioning & Cost Complete. Floor Layouts, Sectional Elevations, and Furniture Layout of the Different Interior Spaces. Detail layout plan (/tablet plan) of individual space/unit, with 4 sectional elevations and detailed specifications. Design and detailing of furniture and interiors according to consumer ergonomics Application of Colour, Texture &. Pattern through Techniques of Rendering & Presentation. Visual and oral presentation of complete project.	7	21
IV	**Introduction to and Familiarization with Sketchup:** 1.Interface and basic commands: Introduction to the interface, setting up a drawing template, familiarization with large tool set, scenes, importing .dwg files, saving files, installing extensions, etc. 2. Drawing and Modifying: Drawing and Modifying objects,	7	21

	line, rectangle, circle, arc, object properties, Units, dimensions, lines, and pen weight. Blocks, group, extrude, rotate, add texture, etc. 3.Presentation and Render: Learning analysis of a space to represent it as a final 3D model with renders or walkthrough. Project: Students to be asked to create a 3D model of the residence in Sketchup and Render the output in software.		
TOTAL		**112**	

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	28	84	8 hours
			Group Work Presentation/ Report Making Site Visits

Text Books:

1. Booth, Wayne C., Colomb, Gregory G. and Williams, Joseph M. (2003), "*The Craft of Research*", Second Edition. Publishers: The University of Chicago Press, Chicago.
2. Vaux, D.E., & Wang, D. (Eds.). (2020). Research Methods for Interior Design: Applying Interiority (1st ed.). Routledge. https://doi.org/10.4324/9780429029325

Reference Books:

1. Sommer, Robert & Sommer, Barbara (2002), "A Practical Guide to Behavioural Research: Tools and Techniques". New York: The Oxford University Press.
2. Creswell, John W. (2003), "*Research Design: Qualitative, Quantitative, and Mixed Method Approaches*". 2nd Edition. Thousand Oaks, CA: Sage Publications.
3. Borden, I. and Ray, K. R. (2006), "*The Dissertation: An Architecture Student's Handbook.* Second Edition. Oxford: Architectural Press.
4. Anderson, J. and Poole, M. (1998), "*Thesis and Assignment Writing*". Brisbane: John Wiley